

The Importance of Being Even: Restitution and Cooperation*

Maria Bigoni¹, Marco Casari¹, Andrea Salvanti²,
Andrzej Skrzypacz³, and Giancarlo Spagnolo⁴

¹University of Bologna

²University Pompeu Fabra and BSE

³Stanford GSB

⁴University of Rome Tor Vergata & EIEF

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Abstract

We study – empirically and theoretically – how restitution helps restore cooperation after a defection. Restitution strategies propose a return to cooperation by cooperating against defection, and condition actions on the balance between cooperation given and received. We reanalyze experimental data from repeated Prisoner’s Dilemma games and find empirical support for restitution strategies in general, and for a strategy we name Payback, in particular. Besides explaining how subjects deal with conflicts, accounting for restitution strategies helps reconcile discrepancies between theory and experiments—such as the widespread use of non-equilibrium strategies like Tit-for-Tat and the limited predictive power of risk dominance under imperfect monitoring.

Keywords: Experiments, Repeated games, Restorative justice, Social dilemmas, Strategic risk.

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1 Introduction

Cooperation is crucial to many economic and social relationships, and may be fragile or easily disrupted. When trust is breached – whether deliberately or not – rebuilding mutual confidence is essential. We focus on strategies for restoring cooperation after a defection, and argue that *restitution* is a particularly intuitive and effective approach when communication is absent. Restitution entails “proposing” a return to cooperation by restoring the gains lost in the breach. This gesture, which is materially costly for the offender and rewarding for the offended party, may serve as a credible message.

Restitution can also be seen as fair, as it closes the payoff gap created by the deviation and makes subjects even. It constitutes the basis of “corrective” or “restorative” justice—the foundation of modern tort law—as formulated, for example, by Aristotle.¹ It also underpinned many informal decentralized legal systems, from the American Indian, the Australian Aboriginal and the Sardinian Babagian honor codes, to the Anglo-Saxon and Germanic people feuds, where restitution-based rules led to swift reconciliation, avoiding prolonged conflicts or escalation (Jacob, 1977).² Restitution-based restorative justice approaches are now increasingly adopted for juvenile crimes because of their capacity to restore harmony between victims and offenders (Adukia et al., 2025).

Beyond re-establishing cooperation after a breach, restitution may mitigate the effects of strategic risk. Subjects willing to cooperate but expecting their opponent to defect with high probability may begin with an exploratory defection. If the opponent instead cooperates, restitution enables them to revert to mutual cooperation – provided the opponent is playing a forgiving strategy.

While the potential role of restitution in rebuilding cooperation after a breach has been discussed in theoretical studies, it attracted little attention in the experimental literature on repeated games. Its exploration role, by contrast, seems to have been largely absent from both theoretical and experimental work. In this study, we examine the presence and role played by restitution strategies in experimental data from three broad classes of repeated games: finitely and indefinitely repeated Prisoner’s Dilemmas with perfect monitoring, and indefinitely repeated games with imperfect monitoring.

¹ He writes “*For here the law looks only at the difference that stems from the harm done, and it treats persons as equals: if the one person acts unjustly, the other suffers injustice; and if the one did harm, the other was harmed. As a result, since the unjust in this sense is an inequality, the judge tries to restore equality*”. (Nicomachean Ethics, book 5 ch.4)

² The Anglo-Saxon and Germanic feud rules included ‘Wergild’ (Man-Price), a well defined monetary price due by a perpetrator to the family/clan of a victim of homicide, instead of revenge, function of the “value” of the victim (free man, noble...). Weber (Economy and Society, 1921) viewed the feud as a “pre-legal form of order” — a legitimate legal institution within a society lacking a formal centralized monopoly on violence.

We define restitution strategies as a class of conditionally cooperative strategies that respond to unresolved imbalances generated by past play. They are conditionally cooperative in that they prescribe cooperation in at least one state and punish – by reducing cooperation – following a unilateral defection that breaches mutual cooperation. Their crucial characteristic, however, is that imbalances generated by past play affect the strategy’s cooperation prescriptions, and that these imbalances can be settled: when a unilateral defection disrupts mutual cooperation, generating a “debt”, a subsequent compensating unilateral cooperative action can restore the relationship to the same behavioral state as before the breach. Once settlement occurs, the earlier asymmetry no longer affects future play.

To study the role played by restitution strategies in experimental data, we rely on three meta-datasets, each covering a distinct class of repeated games. One includes indefinitely repeated games with perfect monitoring (Dal Bó and Fréchette, 2018); another comprises finitely repeated games with perfect monitoring (Embrey et al., 2017); and a third – which we constructed ex novo – features indefinitely repeated games with imperfect monitoring (Fudenberg et al., 2012; Arechar et al., 2017; Aoyagi et al., 2019).

In all three datasets, we find that the frequency of cooperation depends not only on the outcome of the previous round of interaction but also on the round before that. Consider, for instance, participants’ behavior following a unilateral defection by their opponent: cooperation is more frequent when, in the round before that defection, the opponent had cooperated *and* the participant defected – increasing the imbalance – compared to any of the other possible histories. This finding aligns with the use of restitution strategies – where the original defector allows the opponent to take vengeance and get even before returning to cooperation – and is difficult to reconcile with the strategies most commonly identified in the earlier experimental literature, including Grim trigger (Grim), Tit-for-Tat (TFT), t -period punishment strategies, and their more lenient and forgiving variants. It also suggests that subjects often rely on strategies with memory longer than one even with perfect monitoring. In fact, our analyses indicate that restitution can occur after more than two periods: subjects are more likely to cooperate after a unilateral defection of the opponent when, based on earlier outcomes, they were ‘in debt’ to the opponent at the time the opponent defected.

Restitution strategies tend to balance the number of cooperative actions by the subject and her opponent – often leaving them roughly even. Their adoption may therefore reflect a heuristic rooted in norms of justice, fairness and inequity-aversion (Fehr and Schmidt, 1999; Charness and Rabin, 2002), rather than, or alongside, strategic considerations. This may help explain why we find evidence consistent with the adoption of restitution strategies also in finitely repeated games, even though they cannot be equilibrium strategies in this setting.

To deepen our understanding of the role played by restitution strategies, we then narrow our focus to a specific, simple one that has featured prominently in the evolutionary game theory literature, where it is referred to as “Sophisticated” or “Contrite” Tit-for-Tat. Here

we re-name it *Payback* to emphasize its restitution properties and the contrast with its main “rival” TFT. Payback differs from TFT off the equilibrium path. Consider a period t in which both players cooperate, and a period $t + 1$, in which the player defects — maybe because of an error — and the opponent cooperates. The prescribed action for the player in $t + 2$ is cooperation for both the Payback and TFT strategies. However, if the opponent defects in $t + 2$, then the prescribed action in $t + 3$ diverge for the two strategies. Payback prescribes cooperation, because its debt, which was created by its own defection in $t + 1$, has been repaid, whereas TFT prescribes defection in response to the opponent’s defection in $t + 2$. What happens next depends on the opponent’s strategy. For instance, if the opponent plays TFT, then playing Payback restores mutual cooperation after a single round of punishment by the opponent, while instead playing TFT starts cycles of cooperation and retaliation.³

We study the theoretical and empirical properties of Payback, compared to Grim and TFT, in infinitely repeated games where cooperation can be sustained in equilibrium. Under perfect monitoring, Payback can form a subgame perfect Nash equilibrium (SPNE) when TFT cannot. Moreover, when the gains from unilateral deviations are equal to the losses from playing cooperation while the opponent defects, Payback constitutes an optimal penal code in the sense of Abreu (1988): it sustains cooperation as a SPNE for the same set of parameters as Grim, and is not subject to renegotiation concerns – in line with Van Damme (1989).

Games with imperfect public monitoring are particularly relevant in this context, as the differences between Payback and other common strategies – such as Grim and TFT – emerge more prominently in these settings. Under imperfect monitoring, deviations occur on the equilibrium path, making the ability to swiftly restore cooperation after deviations especially valuable (Fudenberg et al., 2012). We show that, in such environments, Payback (i) can be part of a perfect public equilibrium; (ii) is an equilibrium for a larger set of parameters than Grim; (iii) achieves a higher value on the equilibrium path; and (iv) is less exposed to strategic risk than either Grim or TFT.

The theoretical advantages of Payback over TFT and Grim led us to wonder how often subjects adopt this specific restitution strategy. To address this question, we estimate strategy frequencies using maximum likelihood (Dal Bó & Fréchette, 2011), with and without Payback included in the candidate set. In so doing, we also consider the suspicious variant – *Suspicious Payback* (S-Payback) – which begins with a defection, and “becomes” Payback after the opponent cooperates. Although S-Payback does not constitute a symmetric subgame-perfect equilibrium, it captures the exploratory role of restitution for subjects who would conditionally cooperate, but hold pessimistic beliefs about their opponent’s strategy.

Looking at the last four supergames, we find that in indefinitely repeated games, Payback and S-Payback are played by a share of subjects comparable to that of TFT and its suspi-

³ A formal definition is in Appendix A.

cious variant S-TFT. By contrast, Payback and S-Payback seem to play a less prominent role with finite horizon, as might be expected given their longer memory and the limited scope for future interaction in this setting. Introducing Payback and S-Payback into the estimated strategy set primarily reduces the share attributed to TFT and S-TFT in the indefinite treatments, under both perfect and imperfect monitoring. In other words, omitting Payback and S-Payback leads to TFT and S-TFT being overrepresented in the estimated strategy distribution. This reinterpretation may explain the sustained cooperation observed in treatments where, unlike Payback, TFT (and sometimes Grim) cannot support cooperation in equilibrium. It also offers a fresh perspective on the numerous earlier experiments in which TFT appeared to play a central role in classifying behavior, despite not being an equilibrium strategy.

Our theoretical analysis also yields empirical implications for our understanding of the determinants of cooperation, particularly in environments with imperfect monitoring. Fudenberg et al. (2012, p. 733) noted that their data “do not show the strong support for risk dominance of *TFT* as the key determinant of the level of cooperation in games with noise that was seen in studies of games without noise.”⁴ We revisit this issue by focusing on Payback. In contrast to TFT, we find that the risk dominance of *Payback* is a strong predictor of cooperation rates: across treatments with imperfect monitoring, cooperation is nearly 20 percentage points higher when Payback is risk-dominant than when it is not.

These findings invite a partial re-evaluation of traditional theoretical tools—such as perfect public equilibrium and risk dominance—that had appeared to lose empirical traction in light of the observed relevance of non-equilibrium strategies such as Tit-for-Tat. By highlighting the role of restitution strategies, our findings help reconcile theoretical and experimental perspectives. More broadly, our findings shed new light on mechanisms that support long-term cooperation. While repeated games in the lab typically span only a few rounds, real-world interactions unfold over longer horizons, where defections—whether deliberate or accidental—are often inevitable. Understanding how cooperation is restored after such breakdowns appears to us crucial to explaining how it can be sustained over time.

The remainder of the paper is organized as follows. After a brief overview of the related literature in Section 2, we move to the empirical analysis. Using meta-data from previous experiments involving repeated Prisoner’s Dilemma games, Section 3 provides evidence for the use of a broad class of restitution strategies. Section 4 introduces Payback, which is one of the simplest restitution strategies, and derives theoretical results characterizing its equilibrium and strategic-risk properties. Section 5 presents results from a maximum likelihood estimation of strategies, providing empirical support for the specific use of Payback and its suspicious version and for the predictive role of risk dominance under imperfect monitoring. Section 6 concludes.

⁴ Emphasis on “of TFT” added for clarity.

2 Related literature

Our paper primarily contributes to the experimental literature on the determinants of cooperation with focus on subjects’ strategy adoption in repeated games (see, for example, Dal Bó and Fréchette (2011), Breitmoser (2015), Dal Bó and Fréchette (2019), and Romero and Rosokha (2023) for the case of perfect monitoring; Fudenberg et al. (2012) and Embrey et al. (2013) for imperfect monitoring; and Friedman and Oprea (2012) and Embrey et al. (2017) for finite horizon).⁵ While previous work has identified a variety of strategies across different environments, little attention has been paid to restitution strategies. The only exception we are aware of is Embrey et al. (2013)’s study of a repeated partnership game, where a family of strategies emerges – Sum and SumN – that fits our definition of a restitution strategy. We add to this literature by documenting the importance of restitution strategies in re-establishing cooperation after defections both under perfect and imperfect monitoring, and showing that their omission may have led to an overestimation of the role of memory-one strategies like TFT.

Our work also relates to Fudenberg and Karreskog (2024), who emphasize the role of early-round behavior in shaping cooperation, and show that reinforcement from past experience drives the adoption of defection even in games where cooperation is sustainable. Our findings complement this perspective: S-Payback offers a natural representation of the behavior of subjects who start with defection but transition to cooperation when met with trust, thus capturing a cautious but adaptable form of strategic learning.

Our analysis also speaks to the literature on strategic risk and equilibrium selection in repeated games (see, among others, Blonski et al. (2011), Dal Bó and Fréchette (2011), Fudenberg et al. (2012), Blonski and Spagnolo (2015), Dal Bó and Fréchette (2018), Ghidoni and Suetens (2022), Kartal and Müller (2024), Boczoń et al. (2025), and Martinez-Martinez and Normann (2025)). In particular, we find that the risk dominance of Payback is a strong predictor of cooperation rates even under imperfect monitoring, where previous studies had cast doubt on the relevance of this concept in noisy environments.

Restitution strategies imply a return to cooperation after a punishment phase characterized by asymmetric play. Their theoretical advantages have been highlighted in a number of contexts, including renegotiation-proofness in the repeated Prisoner’s Dilemma (Van Damme, 1989); repeated games with monetary payments, such as relational contracts (Levin, 2003); tacit collusion in repeated auctions (Skrzypacz and Hopenhayn, 2004); and the folk theorem under imperfect monitoring (Fudenberg et al., 1994).⁶ Renegotiation strategies

⁵ Dal Bó and Fréchette (2018) and Mengel (2018) offer excellent meta-analyses.

⁶ In the realm of pricing games, the superiority of asymmetric punishment strategies over strongly symmetric ones has been established by Athey and Bagwell (2001) and Harrington and Skrzypacz (2007). Asymmetric punishment strategies are not merely theoretical constructs: Harrington (2006), for example, describes several instances of their use by real-life cartels.

have also featured in evolutionary game theory, where Contrite or Sophisticated Tit-for-Tat - that we renamed “Payback” - was praised for its resilience in noisy environments. Sugden (1986) showed that while no pure strategy is evolutionary stable in the standard Prisoner’s Dilemma without noise, mistakes in actions allow the restitution strategy Sophisticated Tit-for-Tat to be evolutionary stable. This seminal contribution was further expanded in several subsequent works (Boyd, 1989; Wu & Axelrod, 1995; Boerlijst et al., 1997; Graser & van Veelen, 2024; Mathew & Boyd, 2026). In this paper we reconcile these theoretical results with experimental evidence showing that these strategies are frequently adopted in the lab.

The spirit of restitution strategies also echoes findings in the literature on fairness and inequity aversion, which highlights the role of reciprocity and balance – rather than purely strategic reasoning – in shaping behavior (Fehr & Schmidt, 1999; Bolton & Ockenfels, 2000; Charness & Rabin, 2002). Restitution strategies, which tend to equalize payoffs, may appeal to fairness-related heuristics. The role of restitution in repairing cooperation also links to research on apologies and trust restoration (see, e.g., Halperin et al., 2022). While such studies typically examine verbal or monetary signals, restitution strategies act as costly behavioral analogues.

Finally, we relate to the growing experimental literature on beliefs in repeated games. Recent work shows that strategy adoption is shaped by expectations about others’ behavior. Aoyagi et al. (2024) find that subjects tend to underestimate their opponent’s cooperativeness in the early rounds of indefinitely repeated Prisoner’s Dilemma games, even after a history of mutual cooperation. Similarly, Gill and Rosokha (2024) find that initial beliefs strongly predict strategy choices, and that many subjects with pessimistic beliefs adopt uncooperative strategies despite being otherwise inclined to conditionally cooperate. This makes the exploratory use of restitution strategies – particularly in their suspicious versions – an attractive option.

3 Reduced-form evidence on restitution strategies

In this section, we present reduced-form empirical results obtained using data from previous experiments involving repeated Prisoner’s Dilemma games. In Appendix A we provide the precise definition of restitution strategies which informs the subsequent analyses.

We utilize data from three different meta-datasets, which we refer to as STANDARD, NOISE, and FINITE, as detailed in Table 1. Each of these datasets features the Prisoner’s Dilemma as the stage game and is structured into sequences of play involving fixed pairs of participants, which we term ‘supergames’. However, they differ in the time-horizon of the supergame (which is either indefinite or finite), and in the level of information provided. Each dataset represents a distinct class of games.

Repeated interaction: Monitoring:	Indefinite Perfect	Indefinite Imperfect	Finite Perfect
Source:	Dal Bo & Frechette (2018)	This paper	Embrey et al.(2018)
Number of sessions:	103	27	27
Number of subjects:	1734	598	552
Number of observations:	116,644	60,334	65,720
Number of papers:	12	3	6
Dataset:	STANDARD	NOISE	FINITE

Notes: Since the analyses we conduct are not compatible with supergames lasting less than 3 periods, we drop those observations from all the datasets. The NOISE dataset includes data from Fudenberg et al. (2012), Arechar et al. (2017) (only the treatments without communication), and Aoyagi et al. (2019) (only the treatments strategically equivalent to the setting of Fudenberg et al., 2012 and Arechar et al., 2017).

Table 1: Summary information on the meta-dataset of repeated PD experiments.

STANDARD includes data from previous experiments on indefinitely repeated games with perfect monitoring: the stage game is repeated with constant continuation probability δ , and subjects observe each other’s action without any noise. The FINITE dataset consists of finitely repeated games with perfect monitoring; hence the stage game is repeated with probability one for a finite number of periods, which is common knowledge. The NOISE dataset encompasses indefinitely repeated games with imperfect monitoring, studied in Fudenberg et al. (2012), Arechar et al. (2017), and Aoyagi et al. (2019). In each period there is a probability E that the realized action or a payoff-relevant signal of the chosen action deviates from the player’s intended action. This deviation occurs independently across the two players. When a deviation occurs, players observe the realized public signals — which determine payoffs — but not the opponent’s intended action. Each player knows whether their own action was exogenously altered, but cannot tell whether this was also the case for the opponent. To take noise into account, the standardization of payoffs reported in Table B1 is done on the expected stage-game payoffs.

3.1 Evidence of restitution strategies

To cleanly isolate restitution strategies from reputational dynamics arising in repeated interactions, in this section we restrict our analysis to decisions made in the third period of each supergame. We focus on instances where the opponent defected unilaterally in period 2. We then analyze how a player i ’s action in period 3 varies depending on the outcome of period 1, that is, with their imbalance in period 2. In this section, we consider the case in which the imbalance is defined as the difference in the number of past unilateral defections. When the imbalance is positive the player is in “debt”, while when the imbalance is negative

the player is in “credit”.⁷

For instance, a sequence of outcomes DD, CD or CC, CD results in a “credit” for player i at the beginning of period 3, because player i has defected once less than the opponent in the supergame. In this case, a restitution strategy prescribes a higher likelihood of defection for player i , while an opponent who is also following a restitution strategy should be more likely to cooperate, since she has to “repay” the debt. On the other hand, a sequence of outcomes DC, CD settles the imbalance as both players have cooperated an equal number of times. In this case, a restitution strategy prescribes a higher likelihood of cooperation for player i than in the previous example. If the opponent was initially cooperating, then the compensating sequence settles the imbalance and, by our definition, also restores the original behavioral state.

This latter scenario is particularly relevant in our analyses, because it serves as a litmus test: in it, restitution strategies diverge from other common strategies such as Tit-for-Tat, Grim-Trigger, or t -period punishment. While restitution strategies align with the above strategies in most other cases, this particular instance allows us to tell them apart.

Table 2 presents the empirical frequency of cooperation in period 3, following CD outcome in period 2. This is done for all possible outcomes in period 1 (DC, DD, CD and CC). The number of observations for each history is also provided in parentheses.

Finding A. *Consistent with restitution strategies, in the third period of a supergame subjects cooperate more often following a unilateral defection by the opponent in the second period, if they themselves have unilaterally defected in the first period.*

Support for Finding A emerges from Table 2 which shows that cooperation rates in the DC, CD history are approximately 57%, compared to the 24%-30% observed in the other three cases of the STANDARD dataset. This is the pattern predicted by restitution strategies, in contrast to Grim or TFT that, after CD, predict defection (DD or DC) and no variation with period 1 outcome (no differences across rows).

A similar pattern is observed in the FINITE dataset, with a cooperation rate of 52% versus 22%-25% in other cases. In the NOISE dataset, we categorize observations based on the intended action in period 1. This provides insight into the heterogeneity of public strategies adopted by subjects, because the first action is not affected by the history (though it may depend on learning in previous supergames, Fudenberg and Karreskog, 2024). The pattern in the NOISE dataset mirrors those in the other datasets under both conditions.

⁷ This scalar index is a canonical selection, not an implication of the axioms presented in Appendix A. Hence, the above specification – which is linear in the imbalance – tests the axioms exactly for the balance-measurable subclass of restitution strategies (the “Sum” family of strategies in Embrey et al., 2013), while it does so only directionally for the full class.

Outcome in period 1	Imbalance at start of period 2	STANDARD	NOISE		FINITE
			Init.intention C	Init.intention D	
DC	debt	0.566 (746)	0.859 (99)	0.475 (198)	0.522 (458)
DD	even	0.282 (440)	0.521 (73)	0.186 (167)	0.242 (132)
CD	credit	0.297 (617)	0.240 (425)	0.179 (28)	0.246 (284)
CC	even	0.239 (268)	0.632 (353)	0.100 (10)	0.217 (138)
Total		0.383 (2071)	0.472 (950)	0.325 (403)	0.367 (1012)

Notes: The imbalance reflects the difference between the player and the opponent in the total number of realized defections, with a positive value indicating a debt. In the NOISE dataset, the initial intention is the action chosen by the subject in *period 1* before the random realization of the noise and the cooperation rate, here and in the forthcoming tables, is computed using the intended action. Number of observations in parentheses.

Table 2: Cooperation frequency in period 3 following a CD outcome in period 2.

Finding A is supported by the panel regressions reported in Table 3, which include individual and supergame fixed effects and cluster standard errors at the individual level.

The large and highly significant coefficient for the DC outcome in period 1, observed across all regressions, suggests that restitution strategies are widely adopted to reestablish cooperation in repeated social dilemmas. Notice that the results in Tables 2 and 3 are also incompatible with lenient strategies such as Tit-for-two-tats: these strategies would predict the same cooperation rate when the outcome in period 1 is CC or DC, whereas we observe a significantly higher rate when the initial outcome is DC.

3.2 Beyond the first three periods

While focusing on the first three periods offers clean evidence for repayment strategies following an initial defection, it excludes repayment behavior after initial cooperation. In this sense, the estimates obtained are a tight lower bound. This section extends our analyses to the following periods to gain further insights on more general restitution behavior.

Table 4 reports the results of regressions in which the dependent variable is the subject's action in period t , conditional on the outcome in period $t - 1$ being CD. The independent variable is the imbalance at the beginning of period $t - 1$ when the opponent's unilateral defection took place. The results indicate a significant correlation between the imbalance and cooperation in periods following a unilateral deviation by the opponent, in all the three classes of experiments we consider. Thus, the analysis of data from periods beyond the

	STANDARD	NOISE		FINITE
		initial C	initial D	
DC (debt)	0.370*** (0.056)	0.141*** (0.054)	0.355** (0.148)	0.363*** (0.071)
DD (even)	0.060 (0.060)	-0.275*** (0.075)	0.199 (0.146)	0.053 (0.083)
CD (credit)	-0.051 (0.057)	-0.497*** (0.041)	-0.061 (0.151)	-0.083 (0.072)
Intercept	0.264*** (0.071)	0.722*** (0.056)	-0.062 (0.183)	0.210*** (0.081)
N. of observations	2071	950	403	1012
N. of subjects	949	394	244	348
R ² -within	0.173	0.331	0.156	0.196
R ² -between	0.036	0.107	0.045	0.061
R ² -overall	0.081	0.196	0.071	0.113

Notes: Linear probability model with subject fixed effects and supergame fixed effects, based on one observation per subject, per supergame. The omitted category is outcome CC in period 1. Standard errors clustered at the subject level. *** $p < .01$, ** $p < .05$, * $p < .1$.

Table 3 Cooperation in period 3 when the outcome was CD in period 2.

third one confirms Finding A, showing that subjects react to a unilateral defection by the opponent in a different way, depending on the imbalance when the defection took place.

One might wonder whether the situation of being “in debt” simply captures the outcome in $t - 2$, or instead subjects also react to “debts” that were generated earlier in time. In fact, restitution strategies predict higher cooperation with a “debt”, even considering histories before $t - 2$. Here we propose a method to ascertain the empirical relevance of restitution strategies that employ longer memory spans. We replicate the regression presented in Table 4, restricting the sample to interactions where the outcome in $t - 2$ was DC. If subjects only reacted to debts that emerged in $t - 2$, then the coefficient of the variable Imbalance_{t-1} should no longer be significant. The results of this analysis are presented in Table 5. They reveal that even if the imbalance in cooperation emerged more than two periods before, the probability of cooperation is still higher when subjects are in debt. The effect, however, is only significant in STANDARD, and in NOISE when the initial intention was cooperative. This is broadly in line with the use of restitution strategies with memory longer than two periods. These results are consistent across all classes of games, and contrast with the predictions of both Grim trigger and Tit-for-Tat strategies, which prescribe defection after a CD outcome, regardless of the history of play before period $t - 1$.

Finding B. *When considering all periods in a supergame, empirical evidence consistently aligns with restitution strategies, documenting the relevance of strategies that employ a memory span of two or more periods both with perfect and imperfect monitoring.*

	STANDARD	NOISE		FINITE
		initial C	initial D	
Imbalance _{<i>t</i>-1}	0.127*** (0.011)	0.204*** (0.012)	0.038*** (0.011)	0.225*** (0.035)
Period	0.001 (0.001)	-0.004** (0.002)	-0.001 (0.002)	-0.041*** (0.005)
Imbalance _{<i>t</i>-1} × Period	-0.004*** (0.001)	-0.007*** (0.001)	-0.002*** (0.001)	-0.023*** (0.005)
Intercept	0.295*** (0.024)	0.592*** (0.029)	0.209*** (0.033)	0.570*** (0.038)
Number of observations	6161	5683	2851	3827
Number of subjects	1283	499	378	438
R-squared for within model	0.071	0.150	0.018	0.112
R-squared for between model	0.011	0.029	0.001	0.009
R-squared for overall model	0.001	0.051	0.008	0.038

Notes: Linear probability model, with one observation per subject, per period. Imbalance is measured at the beginning of the previous period and equals the opponent's cumulative realized cooperation minus the subject's cumulative realized cooperation since the beginning of the supergame. The model includes subject fixed effects and supergame fixed effects. Standard errors are clustered at the subject level. *** $p < .01$, ** $p < .05$, * $p < .1$.

Table 4: Cooperation in period t when the outcome was CD in $t - 1$ (all periods after 2).

Although they did not consider restitution strategies, Fudenberg et al. (2012) highlighted the importance of considering strategies with memory longer than 1 with imperfect monitoring. Besides pointing at the role of restitution, Finding B indicates that it is important to consider strategies with memory longer than 1 also with perfect monitoring. Considering these strategies is particularly relevant to understand behavior in histories in which subjects are off the equilibrium path due to pessimistic beliefs and the need for exploration.⁸ The analysis in Appendix D shows that restitution strategies are very effective in complementing other cooperative strategies, ensuring a fast return to mutual cooperation after the exploratory phase, and keeping aggregate cooperation rates substantial, both with perfect and with imperfect monitoring.

The focus on histories that involve miscoordination is crucial for uncovering how individuals reestablish coordination after a breach, as opposed to studying when cooperation emerges in the first place (Dal Bó and Fréchette, 2018). The choices made in these histories have a potentially large impact on cooperation rates, as they shape the outcomes of all subsequent periods; hence the importance of identifying restitution behavior clearly. Although such instances are relatively rare in the data, they are encountered at least once by roughly

⁸ Potential mis-coordination where the outcome in period 1 is CD or DC happens in 37% of the supergames in STANDARD.

	STANDARD	NOISE	FINITE
		initial C	initial D
Imbalance _{<i>t</i>-1}	0.061* (0.032)	0.153*** (0.024)	0.011 (0.031)
Period	0.000 (0.003)	-0.007* (0.004)	-0.001 (0.005)
Imbalance _{<i>t</i>-1} × Period	-0.002 (0.001)	-0.005*** (0.002)	-0.001 (0.002)
Intercept	0.224** (0.088)	0.626*** (0.065)	0.238*** (0.078)
Number of observations	1103	1078	540
Number of subjects	565	388	245
R-squared for within model	0.087	0.115	0.086
R-squared for between model	0.000	0.000	0.003
R-squared for overall model	0.009	0.024	0.021

Notes: Linear probability model, with one observation per subject, per period. Imbalance is measured at the beginning of the previous period and equals the opponent’s cumulative realized cooperation minus the subject’s cumulative realized cooperation since the beginning of the supergame. The model includes subject fixed effects and supergame fixed effects. Standard errors are clustered at the subject level. *** $p < .01$, ** $p < .05$, * $p < .1$.

Table 5: Cooperation in period t when the outcome was CD in $t - 1$ and DC in $t - 2$ (all periods after 3).

three-quarters of subjects.⁹

The reason why the tables in this Section rely on a small subset of observations (7.6%, see Tables 1, 4) is that – even though an overwhelming majority of subjects do encounter situations in which restitution could arise – in almost all other periods the prescriptions of restitution strategies coincide with those of well-known strategies such as Tit-for-Tat. Consequently, even if all participants were following restitution strategies, an observer might mistakenly infer that they were employing Tit-for-Tat (or vice versa). We believe that this is one important reason why restitution strategies have been empirically understudied so far.

3.3 The role of experience

In our three datasets, subjects participated in multiple supergames during each experimental session. To examine the strategy adoption while subjects gain experience, we contrast the empirical patterns that emerged in the earliest supergames with those from the latest supergames within each session.

We adopt the same approach as in Subsection 3.2, and contrast the first and last four supergames of each session. Figure 1 illustrates the frequency of cooperation following a

⁹ About 74% in STANDARD, above 83% in NOISE and 79% in FINITE (no. of subjects in Tables 1, 4).

unilateral defection by the opponent (CD), with Tables 6 and 7 providing additional analyses. We partition observations based on the subject’s imbalance – whether in debt or not – in the period when the opponent unilaterally defected.

	STANDARD	NOISE		FINITE
		initial C	initial D	
Imbalance _{<i>t</i>-1}	0.116*** (0.018)	0.217*** (0.018)	0.099*** (0.023)	0.219*** (0.061)
Period	0.002 (0.002)	-0.005 (0.003)	0.003 (0.005)	-0.045*** (0.008)
Imbalance _{<i>t</i>-1} × Period	-0.003*** (0.001)	-0.007*** (0.001)	-0.003** (0.001)	-0.019** (0.008)
Intercept	0.312*** (0.029)	0.608*** (0.034)	0.151*** (0.043)	0.595*** (0.049)
Number of observations	1907	1840	940	929
Number of subjects	804	430	302	320
R-squared for within model	0.055	0.153	0.038	0.113
R-squared for between model	0.003	0.000	0.018	0.010
R-squared for overall model	0.001	0.036	0.000	0.002

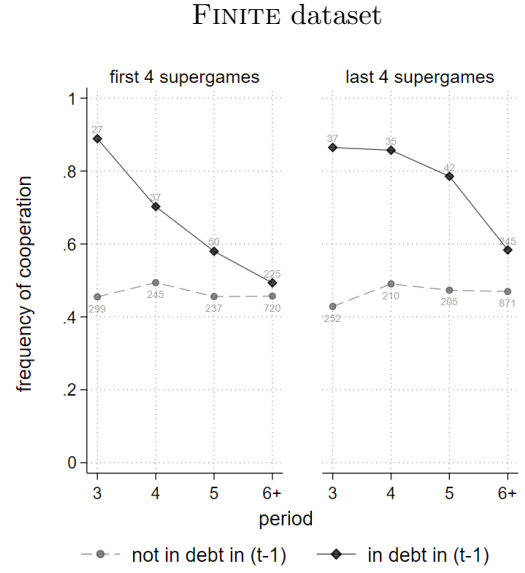
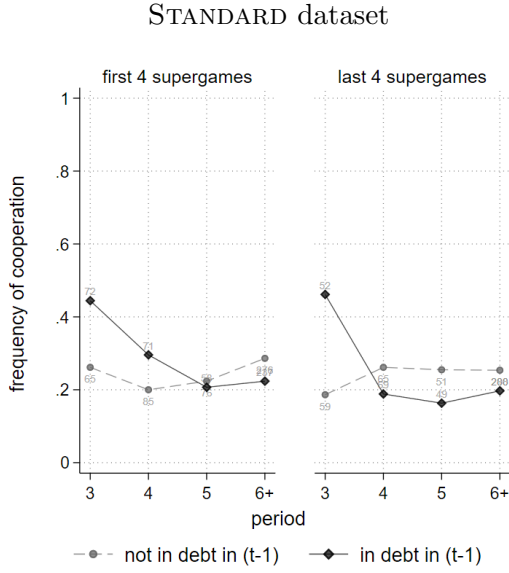
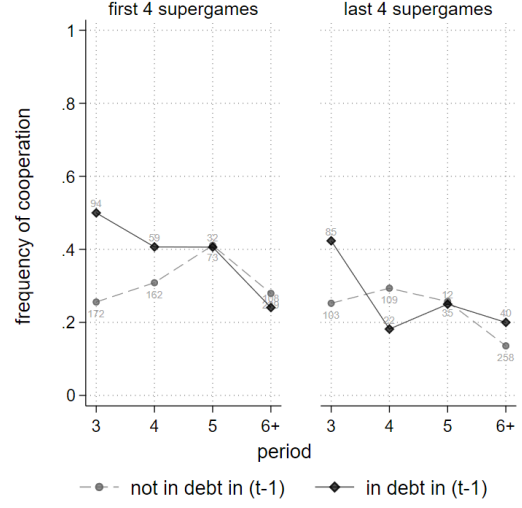
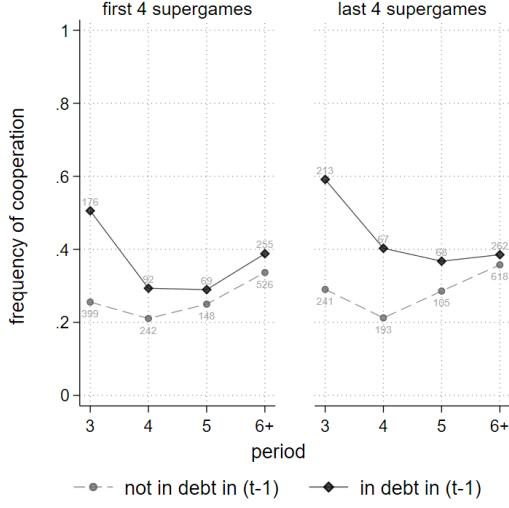
Notes: Linear probability model, with one observation per subject, per period. Imbalance is measured at the beginning of the previous period and equals the opponent’s cumulative realized cooperation minus the subject’s cumulative realized cooperation since the beginning of the supergame. The model includes subject fixed effects and supergame fixed effects. Standard errors are clustered at the subject level. *** $p < .01$, ** $p < .05$, * $p < .1$.

Table 6: Cooperation in period t when the outcome was CD in $t - 1$ (all periods after 2), first four supergames.

Let us point our attention to the first four supergames, represented in Figure 1. The solid line typically lies above the dashed line, suggesting a higher frequency of cooperation among subjects who were in debt when their opponent unilaterally defected, compared to those who were even or in credit. This observation is corroborated by the results in Table 6 and aligns broadly with Findings A and B, which indicated the adoption of a restitution strategy by a sizable fraction of subjects.

Let us now turn to the last four supergames. In the STANDARD and FINITE datasets, the effect not only endures with experience, but it also grows in magnitude. When evaluating the evidence under imperfect monitoring (NOISE dataset), it is useful to account for the initial intention in period 1. With an initial intention to cooperate, the effect persists and marginally increases in size. Instead, if the initial intention is to defect, the effect – already smaller for inexperienced subjects – further shrinks with experience. This difference relative to perfect monitoring may be linked to ‘suspicious’ types not completely trusting the signal of their opponent cooperation.¹⁰

¹⁰ Figure 1 and Tables 6 and 7 also provide a more detailed view of the effect’s magnitude within a



Notes: The numbers next to each marker indicate the number of observations.

Figure 1: Inexperienced vs. experienced behavior: cooperation rates conditional on the imbalance (all periods after 2).

	STANDARD	NOISE		FINITE
		initial C	initial D	
Imbalance _{<i>t</i>-1}	0.125*** (0.022)	0.232*** (0.021)	0.066*** (0.024)	0.314*** (0.075)
Period	0.003 (0.002)	-0.002 (0.002)	-0.003 (0.003)	-0.033** (0.014)
Imbalance _{<i>t</i>-1} × Period	-0.003*** (0.001)	-0.007*** (0.001)	-0.002* (0.001)	-0.024** (0.009)
Intercept	0.512*** (0.153)	0.740*** (0.174)	0.700*** (0.088)	0.382*** (0.093)
Number of observations	1767	1897	833	664
Number of subjects	661	406	252	288
R-squared for within model	0.081	0.178	0.059	0.184
R-squared for between model	0.000	0.005	0.002	0.008
R-squared for overall model	0.000	0.067	0.013	0.000

Notes: Linear probability model, with one observation per subject, per period. Imbalance is measured at the beginning of the previous period and equals the opponent's cumulative realized cooperation minus the subject's cumulative realized cooperation since the beginning of the supergame. The model includes subject fixed effects and supergame fixed effects. Standard errors are clustered at the subject level. *** $p < .01$, ** $p < .05$, * $p < .1$.

Table 7: Cooperation in period t when the outcome was CD in $t - 1$ (all periods after 2), last four supergames.

4 Payback and the theory of infinitely repeated games

To deepen our understanding of the general findings reported earlier, in this section, we consider a specific restitution strategy - Payback - that has been studied in the evolutionary literature under the names of Sophisticated-TFT and Contribute-TFT (Sugden, 1986; Boyd, 1989; Wu and Axelrod, 1995; Boerlijst et al., 1997; Graser and van Veelen, 2024; Mathew and Boyd, 2026). We chose Payback because of its appealing evolutionary properties, and because it belongs to the class of the simplest restitution strategies if complexity is measured as the number of states and transitions.¹¹ After describing this strategy, we characterize its theoretical properties in infinitely repeated games, both under perfect and imperfect monitoring, comparing them with those of Grim and TFT.¹²

Payback captures well the re-coordination role of repayment after cooperation breaks down or after an unintended unilateral defection in the first period. The re-coordination role of Payback is particularly important with imperfect monitoring, where breaches of cooperation are unavoidable. However, because it starts cooperating, Payback does not perform

supergame. Across all game classes, the gap between the solid and the dashed lines reaches its peak in period 3, then diminishes. This pattern further refines Findings A and B.

¹¹ See Oprea (2020) and Salant (2011).

¹² We do not discuss finitely repeated games because Payback can never be an equilibrium there.

the exploration role of repayment. This role is well captured by the Suspicious version of Payback (S-Payback), which begins with an exploratory defection, turning into Payback after the opponent cooperates.¹³ We describe S-Payback, but do not characterize its properties because it is not a symmetric equilibrium strategy. Its adoption is not justified on equilibrium grounds, but rather as a response to strategic uncertainty and pessimistic beliefs on the frequency of subjects always defecting (see the Online Appendix C). However, past the exploration stage S-Payback behaves exactly like Payback, inheriting all its properties. At the end of this section we also briefly discuss the complementarities between S-Payback, Payback, and forgiving strategies such as TFT.

4.1 Payback

Figure 2 describes the Payback automaton, which consists of three states represented by circles and a set of transition rules represented by arrows. The upper-case letter within each circle indicates the action to be taken by the agent in that state: we denote cooperation by “C,” and defection by “D.” The pairs of lower-case letters next to the arrows represent the action pairs that trigger the corresponding transition: “cc” denotes mutual cooperation, “dd” mutual defection, “cd” a unilateral defection by the opponent, and “dc” a unilateral defection by the player. As we discuss later, in the case of a game with perfect monitoring, the actions determining transitions are the actions chosen by the players. In the case of imperfect monitoring, these are actions publicly observed by the agents, which may differ from the actions intended by the players.

Payback prescribes to start at the top-left circle, which we term the “Cooperation state”: hence, the initial action is cooperation (C). The automaton remains in the “Cooperation state” if the observed outcome is “cc” or “dd” (hereafter cc, dd). If the outcome is “cd”, it triggers a transition to the “Punishment state,” in the bottom-left circle: it prescribes to defect (D) until observing cooperation from the opponent (“dc” or “cc”). Finally, if the outcome in the cooperation state is “dc”, Payback triggers a transition to the “Restitution state” (in the upper-right circle) which prescribes cooperation (C), until restitution (“cd”) or mutual cooperation (“cc”) occurs triggering a return to the Cooperation state.

The Payback strategy differs notably from the Grim strategy because it punishes deviations asymmetrically: when player 1 defects (intentionally or not), his continuation payoff goes down. At the same time, the continuation payoff of player 2 goes up. In the continuation game, player 2 is paid back for the bad outcome of this period. In contrast, Grim (and any strongly symmetric strategy, like the ones described in Green and Porter, 1984) penalizes both players when one of them defects.

¹³ Similarly to Payback, this strategy belongs to the class of the simplest restitution strategies that start defecting.

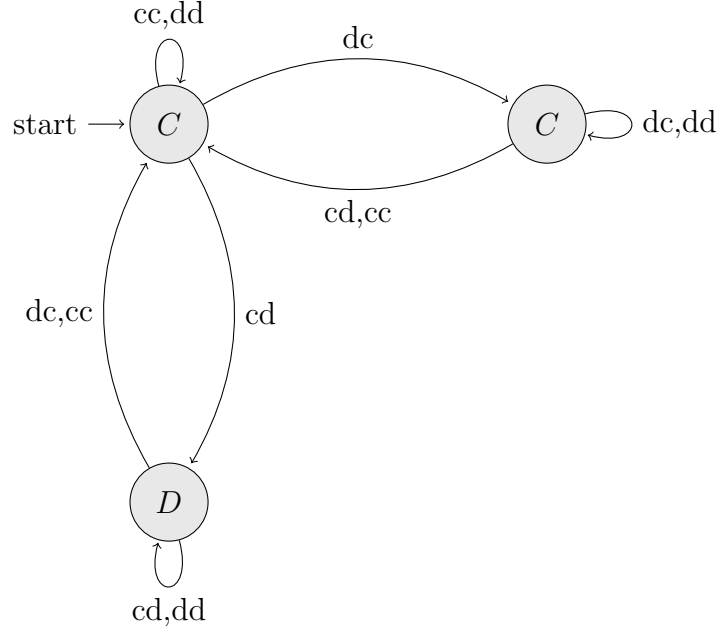


Figure 2. Automaton of “Payback strategy”

At first sight, Payback may look very similar to TFT, but two important differences arise from Payback depending on more than what happened in the last period. In particular, there are two instances that set Payback apart from TFT. First, if the observed outcome is cd , the prescription of Payback depends on the state: cooperation in the Restitution state and defection in the other states. Second, if the agent observes dd , Payback prescribes defection only if the agent is in the Punishment state. Otherwise, it prescribes cooperation. Instead, in both instances, TFT prescribes defection independent of the previous history.

4.1.1 Perfect monitoring

Under perfect monitoring, Grim has the lowest possible critical discount factor (δ) among all strategies that sustain cooperation in every period as a subgame perfect equilibrium (SPE) outcome. The reason is that it is an optimal penal code in the sense of Abreu (1988), as it uses the harshest possible punishment for deviations (in PD, the static Nash payoffs are the min-max payoffs). Hence, while on the cooperative path, the value of cooperation is the same, the critical discount factor for Payback has to be weakly higher than for Grim.

Payback prescribes that a deviation on the equilibrium path, with a gain g on top of the equilibrium payoff, is compensated by repayment in the subsequent period, with an implied loss of l . The larger is g (in comparison to l), the more tempting it is to deviate unilaterally in the cooperative state. Also, if l is too large, there would be an incentive to deviate during the repayment phase. Depending on the parameters, either of these constraints may bind.

We show in Appendix B that the relevant constraint for Payback to be a SPE is:

$$\delta \geq \frac{\max\{l, g\}}{1 + l}.$$

Specifically, if $g > l$, the binding incentive constraint is in the Cooperation state, while if $l > g$, the binding constraint is in the Restitution state. Following this reasoning, in Appendix B we prove the following proposition:¹⁴

Proposition 1. *Consider an infinitely repeated PD with perfect monitoring and normalized payoffs as in Table B1. (i) There exists a threshold $\delta^* < 1$ such that for all discount factors $\delta \geq \delta^*$ Payback is a subgame perfect equilibrium strategy. (ii) If $g = l$, this threshold is the same for Grim and Payback; if $g \neq l$, the threshold is strictly lower for Grim than for Payback. The threshold for risk dominance of Grim is the same as the threshold for risk dominance of Payback $\delta_{RD} = \frac{g+l}{1+g+l}$ for any g and l .*

Note that if $g = l$, then Grim and Payback are SPE for the same set of parameters: the anticipation of the repayment phase offsets gains from a one-sided deviation to the same extent as the commitment to an indefinite punishment. Payback's punishment phase is much more efficient, however, as players can keep cooperating after just one round of punishment (Restitution) in which the defector compensates the opponent for the deviation.¹⁵

Finally, a standard result in the literature is that if $g - l < 1$, as imposed in our game, TFT is not sustainable as an SPE, except in the knife edge case in which $g = l$ and $\delta = \frac{g}{1+g}$. The reason is that after a unilateral deviation, TFT calls for an infinite sequence of mis-coordinated actions (cd-dc) that the players would have incentives to deviate from.

4.1.2 Imperfect monitoring

We now turn to analyzing Perfect Public Equilibria (PPE) of the repeated game with imperfect monitoring. In this subsection, we assume $g = l$ for two reasons. First, in order to theoretically characterize the implications of imperfect monitoring, we want to compare Grim and Payback under a common perfect monitoring benchmark, which by Proposition 1 is

¹⁴ Recall that we have assumed above that $g - l < 1$.

¹⁵ While this does not matter for equilibrium values under perfect monitoring because deviations do not happen on the equilibrium path, it matters in terms of robustness of the strategy to renegotiation. The threat of never cooperating again after a player unilaterally deviates is so inefficient that it may be implausible if players can renegotiate the continuation equilibrium. The punishment phase in Payback is robust to such concerns because it is much more efficient and asymmetric, so carrying out the repayment phase is more profitable for the party that suffered the defection than renegotiating away the punishment phase and immediately returning to mutual cooperation (Van Damme, 1989).

implied by $g = l$. Second, the main related experimental results under imperfect monitoring (Fudenberg et al., 2012; Arechar et al., 2017; Aoyagi et al., 2019) assume $g = l$.¹⁶

The specific game with imperfect monitoring we study has been studied in Fudenberg et al. (2012) (FRD, henceforth) and Arechar et al. (2017). We assume that in each period, players privately choose intended actions. Then, each intended action is changed with probability $E < \frac{1}{2}$, independently across the two players. Players publicly observe the implemented action profile, which is payoff relevant.¹⁷ In Appendix B, we show that in this class of games, Payback is sustainable in a PPE for a larger range of discount factors than Grim:

Proposition 2. *In the infinitely repeated PD game with imperfect monitoring described above, (i) if $\frac{1+g}{g} > \frac{1-E+E^2}{(1-2E)(1-E)}$, then there exist threshold levels for the discount factor, $\delta_{PPE, Payback} < 1$, $\delta_{PPE, Grim} < 1$, above which Payback and Grim are perfect public equilibrium strategies; and (ii) for all $g = l > 0$, and $E \in (0, \frac{1}{2})$, the threshold for Payback is strictly lower than for Grim: $\delta_{PPE, Payback} < \delta_{PPE, Grim}$.*

The intuition for the ranking of the threshold discount factors for Payback and Grim is as follows. Under Grim, cooperation can be sustained in equilibrium for sufficiently high discount factors, but the punishment sustaining cooperation is much more inefficient than for Payback. When players move from mutual cooperation to mutual defection during the Grim punishment phase, the sum of normalized payoffs drops approximately from 2 to 0 (ignoring noise). In contrast, when a player unilaterally defects and then moves to the Restitution state, accepting the opponent’s punishment, the sum of payoffs drops approximately only to 1, while providing even stronger punishment per period (the deviating player loses $1 + l$ instead of 1). So the false-positive punishments caused by noise are just much more efficient (in terms of keeping a high sum of continuation payoffs while satisfying the incentive compatibility constraints) in Payback than in Grim. This improves the expected payoffs from cooperation on the equilibrium path and relaxes the Payback’s IC constraints for a PPE.

While Proposition 2 reveals that Payback is a PPE for a wider range of parameters than Grim, the next proposition shows that, when both are PPEs, Payback also offers a higher expected payoff than Grim:

Proposition 3. *In the infinitely repeated PD game with imperfect monitoring described above, for all $g = l > 0$, $\delta \in (0, 1)$ and $E \in (0, \frac{1}{2})$ such that both Grim and Payback are a*

¹⁶ The reason is that whenever $g = l$ an equivalent representation of the Prisoner’s Dilemma exists, in which cooperation and defection take the “benefit/cost” form, b/c in Fudenberg et al. (2012) notation. Specifically, to cooperate, players have to pay a cost c to give a benefit b to the other player, while defection gives 0 to each party. When $g = l$, the b/c ratio corresponds to $\frac{1+g}{l} = 1 + \frac{1}{g} = 1 + \frac{1}{l}$.

¹⁷ As mentioned earlier, the environment is also strategically equivalent to that in the imperfect public monitoring treatment by Aoyagi et al. (2019), which we also include in our analysis of the metadata.

PPE, the expected payoff in the Payback PPE ($V_{Payback}$) is strictly higher than the expected payoff in the Grim PPE (V_{Grim}).

We know from Fudenberg et al. (2012) that, in our imperfect monitoring game, TFT cannot be sustained in a PPE.¹⁸ In the proof of Proposition 3 we also show that the expected payoff in the Payback PPE ($V_{Payback}$) is higher than the expected payoff from both players following TFT (V_{TFT}) for any $\delta \in (0, 1)$. The reason is again Payback’s ability to keep the value of cooperation high after accidental deviations. Payback can induce a return to cooperation just one period after the accidental deviation (if no other errors occur), while with TFT, a sequence of asymmetric $cd - dc$ outcomes would follow. With TFT, Players would be able to revert to mutual cooperation only after a second random error changes the intended action from d to c .

Strategic risk. Several experiments on the infinitely repeated Prisoner’s Dilemma with perfect monitoring have shown that the “strategic risk” of cooperating, i.e. the expected cost of cooperating when the opponent defects, is a strong predictor of cooperation.¹⁹ Blonski and Spagnolo (2015) and Blonski et al. (2011) showed (theoretically) that a simple measure of such risk can be obtained by restricting attention to the cooperative strategy and the uniquely safe strategy AllD, and then comparing the two by applying Harsanyi and Selten (1988)’s concept of risk dominance to the resulting 2x2 automaton game.²⁰ Dal Bó and Fréchette (2011, 2018) noted that the above comparison can alternatively be performed using the basin of attraction, an easy-to-compute indicator from the theory of dynamic systems. The riskiness of a cooperative strategy is then measured by the size of the basin of attraction of AllD (SizeBAD).²¹ In the Appendix B, we present the calculations for SizeBAD when AllD plays against Payback, Grim, and TFT, and prove the following proposition:

Proposition 4. *In the infinitely repeated PD game with imperfect monitoring described above, for all $g = l > 0$ and $E \in (0, \frac{1}{2})$ strategic risk (SizeBAD) is lower for Payback than for Grim and TFT.*

¹⁸ See Online Appendix D in Fudenberg et al. (2012) for a proof. This result also follows from Graser and van Veelen (2024), who show that, in the presence of errors, Nash equilibrium strategies that can be represented via finite automata must exhibit a structure that they label “self-mirroring”, which immediately rules out Tit-for-Tat.

¹⁹ See, among others, Blonski et al. (2011), Dal Bó and Fréchette (2011), Calford and Oprea (2017), Dal Bó and Fréchette (2018), Ghidoni and Suetens (2022), Boczoń et al. (2025), and Martinez-Martinez and Normann (2025).

²⁰ See Blonski and Spagnolo (2015) for details. The authors also derive a cutoff level for the discount factor, δ_{RD} , below which all cooperation equilibria are risk-dominated by AllD, and cooperation should not be expected. Blonski et al. (2011) derived the same cutoff axiomatically.

²¹ SizeBAD is then the probability that a player must assign to the opponent playing the cooperative strategy in order to be indifferent between playing that strategy rather than AllD. Dal Bó and Fréchette (2018) find that the two measures of strategic risk are empirically indistinguishable in their metadata. Note also that SizeBAD=1/2 when $\delta=\delta_{RD}$.

Under perfect monitoring, without the possibility of a tremble, the value of cooperation of TFT, Grim and Payback is the same, while they all punish indefinitely after the first deviation if matched with AllD. Therefore, the level of strategic risk is the same for all the cooperative strategies considered. Under imperfect monitoring this is not true anymore. On the one hand, Payback loses more against AllD compared to Grim, which in turn condemns AllD to lower profits in a pairwise comparison. On the other hand, due to its forgiving nature, the value of cooperation sustained by Payback in equilibrium is much higher than the value sustained by Grim (Proposition 3). Ultimately, the asymmetric punishment phase of Payback solves the trade-off more efficiently, containing the losses against AllD while keeping the value of cooperation high. Conversely, Grim’s lower exposure to losses against AllD does not outweigh the low efficiency implied by its stricter punishment phase. A similar reasoning, mostly driven by efficiency considerations, holds when we compare TFT with Payback. This implies that, when analyzing repeated games under imperfect monitoring one needs to be careful about which threshold for the discount factor to use. Section 4.2 highlights this point further.

4.2 Suspicious Payback

The equilibrium selection theory based on strategic risk addresses the question whether cooperation will emerge, but is mute on which cooperative strategy will be adopted. Recent studies by Aoyagi et al. (2024) and Gill and Rosokha (2024) highlight that beliefs not only predict cooperation, but also influence strategy choices. Consistent with the literature on strategic risk, they also find that players that adopt Always Defect typically hold negative, inaccurate beliefs and exhibit low trust.

Instead of adopting Always Defect, pessimistic players could “test the waters” by first defecting, then switching to cooperation and following the prescriptions of Payback only after the opponent cooperates.²² This “suspicious” version of Payback (S-Payback) protects against Always Defect and improves on it when some players adopt forgiving cooperative strategies, like Payback or TFT.²³ Thus, S-Payback facilitates a quick return to cooperation after a mismatch, while protecting from strategic risk.

Figure 3 describes the S-Payback automaton. The strategy prescribes to start at the bottom-right circle, which we term the “Exploration state”, with the initial action being defection (D). If the opponent defects and the outcome is “dd”, the pessimistic beliefs that led the

²² Similar strategies have also been analyzed by Graser et al. (2025) within an evolutionary model with the option to leave the partner.

²³ In the Online Appendix C, we show that for certain initial beliefs, Payback and S-Payback can coexist with a fraction of players adopting AllD. The latter can be optimal for players that incorrectly believe that cooperative players starting with C use Grim rather than a forgiving strategy. The reason is that, adopting AllD, they cannot learn the true distribution of strategies.

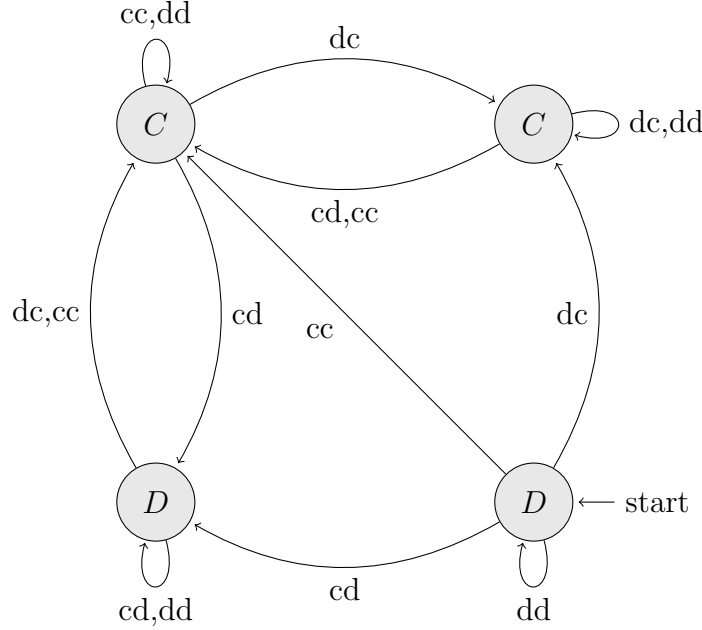


Figure 3. Automaton of “Suspicious Payback strategy”

player to choose S-Payback are validated, prompting continued defection. This is the only case in which the automaton remains in the “Exploration state”. In all the other cases the automaton starts playing like Payback, never going back to the Exploration state. Similarly to Payback and TFT, the main instance setting S-Payback apart from the suspicious version of TFT (S-TFT), which starts defecting and then replicates the opponent’s action in the previous period, is when the observed outcome is “cd”. In this case, the prescription of S-Payback depends on the state: cooperation in the Restitution state and defection in the Punishment state. In contrast, after cd, S-TFT prescribes defection independently of the previous history.

We now highlight two key properties of S-Payback compared to S-TFT when $\delta \geq \delta_{SPE, Payback}$ (this condition encompasses all cases in which repayment is incentive compatible). The reasoning is similar under imperfect monitoring when $\delta \geq \delta_{PPE, Payback}$. First, when Suspicious Payback is matched with Payback, the strategy profile (S-Payback, Payback) forms a Subgame Perfect Equilibrium (SPE)—or a Perfect Public Equilibrium (PPE) under imperfect monitoring—for all subgames not starting from the “Exploration state”. In contrast, the strategy profile (S-TFT, TFT) never constitutes an SPE (or PPE) in the infinitely repeated game. Second, S-Payback exhibits important complementarities with other conditionally cooperative strategies. The reason is that it is better to re-coordinate rather than alternating forever between “dc” and “cd”. Forgiving strategies such as TFT or Payback, matched with S-Payback under perfect monitoring, would receive the payoffs of mutual cooperation starting from period 3. If matched with S-TFT, they would instead receive the

payoff of alternating between “dc” and “cd”. In Figure D1, presented in Online Appendix D, we provide computational analyses that quantitatively illustrate the complementarity of S-Payback and Payback with other strategies.

5 Strategy frequency estimation

This section presents estimates of the frequency with which Payback and S-Payback are adopted in the STANDARD, NOISE, and FINITE datasets (Table 1). We adopt the Strategy Frequency Estimation Method (SFEM) proposed by Dal Bó and Fréchette (2011) which we apply to these two strategies alongside others suggested in the literature. We then examine the empirical role of strategic risk in shaping cooperation rates and strategy adoption under imperfect monitoring, with particular reference to Payback and our Proposition 4.

5.1 Retrieving strategies from observed actions

The evidence from the strategy frequency estimation complements the empirical results presented in Section 3. The SFEM approach differs from the one adopted in Section 3 in at least two respects. First, it explicitly focuses on a single restitution strategy and its suspicious variant, and therefore provides only a lower bound for the role of restitution strategies. Second, it draws on data from all participants and their entire history of play, rather than isolating segments that allow restitution to be distinguished from alternative strategies.

Before estimating the strategies, we restrict the dataset in two ways. First, to mitigate potential learning effects while retaining a sufficient number of observations, we follow the literature (e.g., Dal Bó and Fréchette, 2011) and target the analysis to the final four supergames of each session. Second, on top of implementing the standard approach, which assumes that each participant adopts a single strategy across all supergames, we also divide the dataset into two subsets based on the participant’s initial (intended) action—cooperation or defection. This modification enables a more nuanced understanding of how players re-coordinate after a mismatch, by accurately identifying the proportion of Payback strategies among those that begin with cooperation, and S-Payback strategies among those that begin with defection. While most participants appear in only one subset, some appear in both, having started at least one supergame with each action (27 percent in STANDARD, 25 percent in NOISE, and 19 percent in FINITE).²⁴

²⁴ Table F1 in Online Appendix F provides summary information about the meta-dataset used for the estimation.

Dataset (<i>N. of subjects</i>)		Payback	S-Payback	TFT	S-TFT	ALLD	Grim/GT	Others	Gamma	L.L.
STANDARD										
Full	(1)	0	0	22%	16%	32%	14%	16%	0.36	-10818.94
(1734)	(2)	11%	7%	13%	11%	31%	14%	14%	0.35	-10628.80
Start C	(3)	0	0	47%	0	0	28%	26%	0.31	-4355.68
(1077)	(4)	17%	0	32%	0	0	27%	24%	0.31	-4325.52
Start D	(5)	0	0	0	37%	63%	0	1%	0.32	-4035.73
(1052)	(6)	0	17%	0	21%	61%	0	1%	0.30	-3818.13
NOISE										
Full	(7)	0	0	7%	5%	29%	0	59%	0.48	-7763
(598)	(8)	9%	2%	2%	3%	29%	0	55%	0.47	-7674
Start C	(9)	0	0	14%	0	0	0	86%	0.49	-4802
(430)	(10)	15%	0	5%	0	0	0	80%	0.48	-4733
Start D	(11)	0	0	0	19%	73%	0	8%	0.43	-2306
(286)	(12)	0	15%	0	9%	72%	0	4%	0.43	-2258
FINITE										
Full	(13)	–	–	8%	9%	36%	0	47%	0.37	-3757
(552)	(14)	3%	3%	5%	6%	36%	0	47%	0.36	-3713
Start C	(15)	–	–	15%	0	0	0	85%	0.32	-1845
(363)	(16)	7%	0	8%	0	0	0	85%	0.32	-1844
Start D	(17)	–	–	0	22%	78%	0	1%	0.33	-1068
(327)	(18)	0	8%	0	13%	78%	0	1%	0.31	-1004

Notes: Estimates in rows 2, 8, and 14 are based on the full strategy set, including Payback and S-Payback, whereas rows 1, 7, and 13 exclude these strategies. Rows 4, 10, and 16 restrict the set to strategies that start with cooperation and include Payback, whereas rows 3, 9, and 15 exclude it. Finally, rows 6, 12, and 18 restrict the set to strategies that start with defection and include S-Payback, whereas rows 5, 11, and 17 exclude it. Last two columns: *Gamma* parameterizes the probability of a mistake in the implementation of strategies (equal to $1/(1 + \exp(1/\gamma))$), with a lower gamma implying a higher likelihood that the data arise from the exact adoption of strategies included in the estimation set; *L.L.* (Log Likelihood) indicates the goodness of fit of a model, with a lower L.L. indicating a worse model. The column *Others* includes all strategies considered in Fudenberg et al. (2012) for STANDARD and NOISE (TF2T, TF3T, 2TFT, 2TF2T, Grim2, Grim3), and a different set of strategies for FINITE, following Aoyagi et al. (2024). These include TF2T, Grim2, and a class of threshold strategies that prescribe to conditionally cooperate until a threshold round, then switch to ALLD: Threshold $T-3$, Threshold $T-2$, Threshold $T-1$. The total number of observations is 39,784 for STANDARD, 20,322 for NOISE, and 13,144 for FINITE.

Table 8 Maximum Likelihood Estimates of strategies.

The estimates are computed at the treatment level, where a treatment is defined as a combination of (g, l, δ) , and are then averaged using weights equal to the number of subjects in each treatment. Table 8 reports the SFEM estimates, revealing two main patterns. First, Payback and S-Payback are relatively frequent in STANDARD and NOISE (15%-17%, rows 4, 6, 10, and 12), although they are never the most prevalent strategies. In FINITE, by contrast, their use is much more limited (7%-8%, rows 16 and 18).

Second, when Payback is not included among the estimated strategies, the prevalence of TFT rises substantially (compare row 3 with row 4, and row 9 with row 10). Once Payback is introduced, the frequency of TFT drops by roughly one third in STANDARD and by almost two thirds in NOISE. A similar pattern holds for S-TFT and S-Payback in supergames in which the player begins with defection: the estimated frequency of S-TFT falls between a third and half in both STANDARD and NOISE (rows 5 vs. 6, and 11 vs. 12).²⁵

²⁵ Our focus is on the emergence of repayment behavior, rather than on the prevalence of lenient or

It must be noted, however, that the estimated frequencies of TFT and Payback may lack precision, as only a subset of participants has histories in the last four supergames that allow these strategies to be distinguished. While in NOISE, 60% of participants present at least one round of play that differentiates Payback from TFT, in STANDARD this fraction is only 22%.²⁶ Notably, in STANDARD, about half of participants are classified as either TFT or Payback (Table 8, row 4), yet the evidence available permits us to disentangle the two strategies only for 22% of the participants. For the remaining cases, it is impossible to determine whether they were using Payback or TFT.

5.2 Strategic risk, Cooperation and Strategy Adoption with imperfect monitoring

In this section we re-analyze our NOISE meta-dataset focusing on the effects of strategic risk and risk dominance on cooperation rates.²⁷ As discussed in Section 4.1.2, noise increases the relative appeal of more forgiving strategies that can accommodate accidental deviations without collapsing into permanent defection. In this context, Payback – which retaliates briefly before restoring cooperation – is predicted to perform particularly well. Specifically, with imperfect monitoring, there are parameter ranges in which Payback is a perfect public equilibrium strategy while Grim trigger is not (Proposition 2), and the risk-dominance threshold for Grim trigger and TFT exceeds that of Payback (Proposition 4). Moreover, the expected payoff in the Payback PPE is strictly higher than in the Grim PPE (Proposition 3). The following analysis of the NOISE dataset allows us to assess whether these predictions hold empirically.

To study the relationship between strategic risk and cooperation, different scholars have adopted different conditionally cooperative strategies as benchmarks. For instance, Fudenberg et al. (2012) use TFT as their reference strategy while Dvorak and Fehrler (2023) rely on Grim. Under perfect monitoring, this choice is largely inconsequential. However, as shown in Proposition 4, under imperfect monitoring the threshold at which a cooperative equilibrium becomes risk-dominant depends critically on the reference strategy. In light of this, we partition the NOISE dataset into three groups of treatments, based on the level of strategic risk, using both Grim and Payback as benchmarks. The first group includes

forgiving strategies, which are grouped under the column “Others” in Table 8. Incidentally, the inclusion of Payback has only a modest effect on the estimated frequency of these strategies.

²⁶ In comparison, the issue of lack of precise estimates is less severe for S-Payback versus S-TFT, where the analogous fractions are 56% in STANDARD and 88% in NOISE. The difference between 22% and 56% in STANDARD may explain why the Log Likelihood in row 5 of Table 8 considerably improves in row 6 when introducing S-Payback.

²⁷ Since for the case of perfect monitoring the threshold discount factor for risk-dominance δ_{RD} is the same for Grim and Payback (Proposition 1), the results are mostly aligned with the literature and we report a complementary empirical analysis in Online Appendix E.

treatments with $\delta < \delta_{RD,Payback}$; the second includes those with $\delta_{RD,Payback} \leq \delta < \delta_{RD,Grim}$; and the third includes those with $\delta \geq \delta_{RD,Grim}$. Table 9 reports the cooperation rates and strategy frequencies for each of these three groups.

Level of Strategic Risk	$\delta < \delta_{RD,Payback}$	$\delta_{RD,Payback} \leq \delta < \delta_{RD,Grim}$	$\delta \geq \delta_{RD,Grim}$
Cooperation Rate	27%	42%	61%
Cooperation Rate period 1	45%	62%	73%
Full			
Payback	4%	13%	10%
TFT	7%	0%	1%
S-Payback	3%	0%	3%
S-TFT	6%	5%	1%
AllD	44%	30%	21%
Grim	0%	0%	0%
Others	36%	52%	64%
<i>Observations</i>	4.948	3.992	11.382
<i>Subjects</i>	150	128	320
Begin by C			
Payback	11%	22%	15%
TFT	15%	0%	4%
Grim	0%	0%	0%
Others	74%	78%	82%
<i>Observations</i>	2.200	2.501	8.410
<i>Subjects</i>	86	90	254
Begin by D			
S-Payback	9%	12%	21%
S-TFT	10%	14%	6%
AllD	81%	68%	68%
<i>Observations</i>	2.748	1.491	2.972
<i>Subjects</i>	104	68	114

Notes: The analyses reported in this Table are based on the NOISE dataset. Statistics are based on the last four supergames of each session. All strategies grouped under the row “Others” begin with cooperation. E=probability of error. Estimates of Payback and S-Payback are obtained following the procedures described in Fudenberg et al. (2012).

Table 9 Cooperation Rate and Strategy Adoption in NOISE by Strategic Risk

The first empirical pattern that emerges from Table 9 is a monotonic increase in aggregate cooperation as strategic risk declines: cooperation rates rise from 27% to 42% ($N_1 = 6$, $N_2 = 5$, p-value= 0.10), and then from 42% to 61% ($N_1 = 5$, $N_2 = 10$, p-value< 0.01). A similar trend holds for initial cooperation rates, which increase from 45% to 62% ($N_1 = 6$, $N_2 = 5$, p-value= 0.10) and then from 62% to 73% ($N_1 = 5$, $N_2 = 10$, p-value< 0.05). This pattern stands in contrast to Fudenberg et al. (2012), who conclude that under imperfect monitoring risk-dominance is not such a good predictor of cooperation as it is under perfect monitoring. Using TFT as the reference strategy, they observe that in their set-up TFT becomes risk dominant only in treatments with $g = l = 2/3$, while the largest increase in cooperation occurs between treatments with $g = l = 2$ and $g = l = 1$. On this basis, they conclude that “the risk-dominance criterion has at best limited predictive power regarding

cooperation in games with noise” (Fudenberg et al., 2012, p. 742). Here, we show that this conclusion is reversed when Payback – rather than TFT – is used as a benchmark for risk-dominance. In Fudenberg et al. (2012) data, both initial and aggregate cooperation rates increase significantly from 54% and 32% respectively, in the treatment where Payback is not risk-dominant ($g = l = 2$), to 75% and 49% respectively in the treatment where it is ($g = l = 1$). This resolves the apparent puzzle.²⁸

A second insight from Table 9 is that the estimated adoption of Payback increases significantly when this strategy becomes risk dominant. Among participants who begin the supergame with cooperation, the estimated share using Payback rises from 11% when $\delta < \delta_{RD, Payback}$, to 22% when $\delta_{RD, Payback} \leq \delta < \delta_{RD, Grim}$, before declining slightly to 15% when $\delta \geq \delta_{RD, Grim}$. A similar, and even more pronounced, trend holds for S-Payback among participants who start with defection: its frequency increases from 9% to 12%, and then to 21% across the same thresholds. The non-monotonicity in Payback’s frequency – and the declining prevalence of Grim – merit closer examination. As emphasized by Fudenberg et al. (2012), lenient and forgiving strategies (grouped under “Others”) play a central role in sustaining cooperation under imperfect monitoring. Payback does not compete with these strategies, but rather draws participants away from TFT and Grim. As strategic risk becomes less salient – particularly in the high-cooperation regime where $\delta \geq \delta_{RD, Grim}$ – participants appear more willing to adopt strategies that are slightly more rewarding but also more exposed to strategic risk.

A third notable finding from Table 9 is the shifting balance between Payback and TFT. When $\delta < \delta_{RD, Payback}$, the estimated frequency of Payback is lower than that of TFT (a ratio of about 1 : 1.36). But when $\delta \geq \delta_{RD, Grim}$, the pattern reverses, and Payback becomes substantially more common (ratio 3.75 : 1). A similar shift occurs when comparing S-Payback and S-TFT: their relative frequency moves from 0.9 : 1 to 3.5 : 1 as δ increases. Taken together, these findings suggest that strategic risk is a strong predictor of both cooperation and strategy choice, even in games with imperfect monitoring. Payback emerges as a key catalyst for restoring and sustaining cooperation in environments where actions are implemented with noise.

6 Conclusions

In this paper, we study restitution strategies and their role in restoring cooperation after a voluntary (e.g. exploratory) or erroneous defection, with a focus on both their empirical adoption and their theoretical properties. We rely on meta-datasets comprising many previous experimental studies, covering indefinitely repeated Prisoner’s Dilemmas with perfect

²⁸ Table F3 in Online Appendix F reports our computations for the value of cooperation, the threshold discount factors and the cooperation rates across treatments that differ in the payoff parameters.

and imperfect monitoring, and finitely repeated ones. We find that, in general, restitution strategies are frequently adopted in all three classes of games.

To deepen our understanding of restitution strategies we then narrow our focus to a specific one, Payback, among the simplest in this class. First we theoretically characterize its properties. We show that Payback can support cooperation as an equilibrium outcome in settings with perfect and imperfect monitoring where Tit-for-Tat, and sometimes even Grim trigger, cannot. We also find that, under some circumstances, Payback can be more profitable and less exposed to strategic risk than Grim trigger. Then we use the SFEM method to estimate the frequency of Payback – and of its suspicious version starting with defection – in the three datasets. Considering only the last four supergames to allow for experience, we find evidence of adoption of this specific strategy, particularly with imperfect monitoring, that mainly reduces the estimated adoption rate of Tit-for-Tat.

The combination of our theoretical and empirical analysis also sheds new light on prior findings that, in repeated games with noise, risk dominance does not emerge as a key determinant of cooperation—as it does in the absence of noise. This conclusion stemmed from neglecting restitution-based strategies and the resulting overemphasis on TFT: we find that the risk dominance of Payback is a strong predictor of cooperation rates also in repeated games with imperfect monitoring.

A better understanding of the strategies individuals adopt can also help in designing environments and institutions that make cooperation more resilient to accidental deviations. From this perspective, the pervasiveness of restitution in informal legal systems—and its growing centrality in responses to juvenile crime, where reconciliation between victims and perpetrators is crucial—align well with our findings.

More generally, this study sheds light on factors contributing to the emergence and the long-run resilience of cooperation. While experimental supergames are relatively short, real-world interactions often last so long that defections – whether intentional or unintentional – eventually take place. Identifying how people restore cooperation after a breach of trust therefore appears essential to understanding how cooperation endures over time.

Appendix

A Formal definition of restitution strategies

Let $O = \{cc, cd, dc, dd\}$ denote the stage outcomes of the repeated Prisoner's Dilemma, where the first letter is the player's own (realized, public) action and the second the opponent's. A *history* is a finite sequence $h \in H = \bigcup_{T \geq 0} O^T$, with $\emptyset$ denoting the empty history and $h \cdot o$ (or simply ho) the concatenation of $h \in H$ and $o \in O$; concatenation of histories is defined analogously. A *behavior strategy* is a map $\sigma : H \rightarrow [0, 1]$, interpreted as the probability of cooperating after each history. Under imperfect public monitoring, h is the public history of realized outcomes and σ is a public strategy.

Behavioral equivalence. For $h, h' \in H$, write $h \sim_\sigma h'$ if and only if $\sigma(h \cdot z) = \sigma(h' \cdot z)$ for every $z \in H$ (including $z = \emptyset$). Thus $h \sim_\sigma h'$ when the two histories leave no trace on future play. A set $G \subseteq H$ is $\sim_\sigma$ -closed if $h \in G$ and $h' \sim_\sigma h$ imply $h' \in G$.

Reachability. A history $h = (o_1, \dots, o_T)$ is *reachable* under σ if, for every $t \leq T$, the own-action component of o_t has positive probability under σ given $(o_1, \dots, o_{t-1})$.

The debt order. Let $\succeq$ be the partial order on O generated by

$$dc \succ cc, \quad dc \succ dd, \quad cc \succ cd, \quad dd \succ cd,$$

together with reflexivity; the outcomes cc and dd are unranked. Extend $\succeq$ to equal-length histories coordinate-wise: for $h, h' \in O^T$, $h \succeq h'$ if and only if $o_t \succeq o'_t$ for every $t \leq T$. Thus $h \succeq h'$ means that, period by period, the player has unilaterally defected weakly more, and suffered unilateral defections weakly less, along h than along h' .

Definition 1 (Restitution strategy). *A behavior strategy σ is a restitution strategy if there exists a nonempty, $\sim_\sigma$ -closed set $G \subseteq H$ of reconciled histories, containing at least one reachable history, such that:*

A1 (Stable cooperation) *for every $g \in G$: $\sigma(g) > 0$ and $g \cdot cc \in G$ with $g \cdot cc \sim_\sigma g$;*

A2 (Settlement) *for every $g \in G$: $g \cdot dc \cdot cd \sim_\sigma g$ and $g \cdot cd \cdot dc \sim_\sigma g$;*

A3 (Debt monotonicity) *for every $T \geq 0$ and all $h, h' \in O^T$ with $h \succeq h'$: $\sigma(h) \geq \sigma(h')$;*

A4 (Accountability) *there exists a reachable $g \in G$ with $\sigma(g \cdot cc \cdot cd) < \sigma(g)$.*

B Theoretical analysis of the Payback strategy

To ensure comparability across different studies, we have standardized the payoffs as shown in Table B1. Here, the payoff from deviating from the cooperative action profile (C,C), known as the temptation payoff, is represented as $1 + g$, where parameter g represents the additional gain from defection. Conversely, the loss of payoff from cooperating while the opponent defects, referred to as the sucker's payoff, is denoted as $-l$. Both g and l take strictly positive values.²⁹

		Player 2	
		<i>C</i>	<i>D</i>
Player 1	<i>C</i>	1, 1	$-l, 1 + g$
	<i>D</i>	$1 + g, -l$	0, 0

Table B1: Standardized stage game payoffs for the Prisoner's Dilemma.

B.1 Perfect Monitoring

B.1.1 Proof of Proposition 1

We start by deriving the critical discount factors for the Grim and Payback strategies.

Grim For the Grim strategy, the equilibrium path payoff is $V_C = \frac{1}{1-\delta}$. The only relevant IC constraint is in the cooperative state and Grim is an equilibrium if and only if the deviation payoff in the cooperative state is lower than the on-path payoff, which implies the critical discount factor:

$$1 + g \leq V_C \iff \delta \geq \frac{g}{1 + g}.$$

Payback The Payback strategy has three states: (C, P, R) (Cooperation, Punishment and Restitution). The on-path payoffs in these three states are:

$$\begin{aligned} V_C &= \frac{1}{1 - \delta}, \\ V_P &= 1 + g + \delta V_C, \\ V_R &= -l + \delta V_C. \end{aligned}$$

²⁹ In line with standard practice, we restrict attention to games satisfying $g - l < 1$, which rules out alternating strategies that outperform mutual cooperation. This condition also ensures that a unilateral act of cooperation increases the average payoff — a key property in evolutionary models with assortment, where selection balances individual and group benefits.

The IC constraint in the P state is always satisfied because the player is recommended the dominant stage-game action, and the continuation payoff does not depend on his action. The IC constraint in the C state is

$$1 + g + \delta V_R \leq 1 + \delta V_C \iff \delta \geq \frac{g}{1+l}.$$

The IC constraint in the R state is

$$0 + \delta V_R \leq -l + \delta V_C \iff \delta \geq \frac{l}{1+l}.$$

Combining these two constraints, the threshold discount factor for Payback to be a subgame perfect equilibrium is: $\frac{\max\{l, g\}}{1+l}$. For part (i) note that since $g < 1+l$, the ratio is guaranteed to be less than 1. For part (ii), note that if $g = l$, the threshold discount factors for the two strategies coincide. However, for all $g \neq l$ the threshold discount factor for Payback is higher.

The on path payoff of Grim or Payback when matched with ALLD is $-l$, since they both lose in the first period and get 0 in all subsequent periods. Following a similar reasoning, the on path payoff of AllD when matched with Grim or Payback is $1 + g$. Since there is no difference between Grim and Payback we can compute δ_{RD} as the discount factor that satisfies:

$$\frac{1}{1-\delta} - l = 1 + g \implies \delta_{RD} = \frac{g+l}{1+g+l}.$$

This concludes the proof of Proposition 1. $\square$

B.2 Imperfect Monitoring

Recall that in Section 4.1.2 we assume $g = l$. We claim that whenever $g = l$, an equivalent representation of the Prisoner's Dilemma is the one in table 2b, in which cooperation and defection take the “benefit/cost” form, in which to cooperate, players have to pay a cost c to give a benefit b to the other player, while defection gives 0 to each party.

		Player 2	
		C	D
Player 1	C	1, 1	$-l, 1 + g$
	D	$1 - g, -l$	0, 0

2a) Standardized PD.

		Player 2	
		C	D
Player 1	C	$b - c, b - c$	$-c, b$
	D	$b, -c$	0, 0

2b) Benefit-cost Representation.

Table B2 Equivalence between two representations of a PD.

Specifically, to move from the first representation to the second, substitute $g = l = \frac{c}{b-c}$ and

multiply all the payoffs by $b - c$. We assume $b > c \geq 0$ so that the stage game in this new representation is still a PD.

Recall that we also assume that in every period, each intended action is independently changed with probability $E < \frac{1}{2}$. Denoting $G \equiv \frac{b}{c} = \frac{1+g}{g} > 1$, we can write the stage-game expected payoffs of player 1 as:

		Player 2	
		C	D
Player 1	C	$\frac{c(G-1)(1-E)}{c(G-E(1+G))}$	$\frac{c(E(1+G)-1)}{cE(G-1)}$
	D		

Table B3 Expected Payoffs of Player 1 in the PD with Imperfect Monitoring.

These expected stage-game payoffs constitute a PD because $E \in (0, \frac{1}{2})$ and $G > 1$. Since c scales all expected payoffs equally, without loss of generality, we further normalize $c = 2$ (as in Fudenberg et al., 2012) to describe the stage game with just two parameters (G and E).

B.2.1 Proof of Proposition 2

Payback The expected payoffs in the three states of Payback are, respectively:

$$\begin{aligned}
V_C &= 2(G-1)(1-E) + \delta(V_C(1-E)^2 + (1-E)EV_P + (1-E)EV_R + E^2V_C), \\
V_P &= 2G - 2E(1+G) + \delta(V_C(1-E)^2 + (1-E)EV_C + (1-E)EV_P + E^2V_P), \\
V_R &= 2(E(1+G)-1) + \delta(V_C(1-E)^2 + (1-E)EV_C + (1-E)EV_R + E^2V_R).
\end{aligned}$$

These expressions can be simplified to:

$$\begin{aligned}
V_C &= 2 \frac{(G-1)(1-E)}{(1-\delta)(1+E\delta(1-2E))} \\
V_P &= \frac{2G(1-E) - 2E + \delta(1-E)V_C}{1-\delta E} \\
V_R &= \frac{2(E(1+G)-1) + \delta(1-E)V_C}{1-\delta E}
\end{aligned}$$

To check whether Payback is a (Perfect Public) equilibrium, we need to check the IC constraints in the three states:

Step 1: IC constraints in State C .

A defection in State C gives payoff

$$V_{DC} = 2G - 2E(1 + G) + \delta(V_R(1 - E)^2 + (1 - E)EV_C + (1 - E)EV_C + E^2V_P)$$

For this not to be profitable, we must have

$$V_{DC} \leq V_C$$

The threshold discount factor for state C solves:

$$\begin{aligned} & 2G - 2E(1 + G) + \delta(V_R(1 - E)^2 + (1 - E)EV_C + (1 - E)EV_C + E^2V_P) \\ = & 2(G - 1)(1 - E) + \delta(V_C(1 - E)^2 + (1 - E)EV_P + (1 - E)EV_R + E^2V_C). \end{aligned}$$

The solution is:

$$\delta_C = \frac{2}{V_C - V_R + E(V_P + V_R - 2V_C)} \quad (1)$$

Plugging in the formulas for the payoffs and solving for the threshold δ we get that it is:

$$\delta_C = \frac{-G(1 - 2E)(1 - E) - E + \sqrt{(G(1 - 2E)(1 - E) + E)^2 + 4E(1 - 2E)(1 - E)}}{2E(1 - 2E)(1 - E)}$$

Step 2: IC constraints in State P .

The IC constraint is always satisfied in state P . Note that the transitions from that state are independent of player 1 actions. Hence the continuation payoff of that player does not depend on his action and hence the best response is just myopic best response which is the recommended action in that state.

Step 3: IC constraints in State R .

In state R , the recommended action is C . A deviation to D yields expected payoff:

$$V_{DR} = 2E(G - 1) + \delta(V_R(1 - E) + EV_C).$$

For the deviation not to be profitable, we must have

$$V_{DR} \leq V_R = 2(E(1 + G) - 1) + \delta(V_C(1 - E) + EV_R).$$

The threshold discount factor in that state is:

$$\delta_R = \frac{2}{V_C - V_R}.$$

Recall from (1) that the threshold discount factor in state C satisfies

$$\delta_C = \frac{2}{V_C - V_R + E(V_P + V_R - 2V_C)}.$$

Since

$$V_P + V_R - 2V_C = -2 \frac{(G-1)(1-2E)}{1 + \delta E - 2E^2 \delta} < 0,$$

we have that $\delta_C \geq \delta_R$ for all parameters. Hence Payback is an equilibrium if and only if

$$\delta \geq \delta_{PPE, Payback} \equiv \frac{-G(1-2E)(1-E) - E + \sqrt{(G(1-2E)(1-E) + E)^2 + 4E(1-2E)(1-E)}}{2E(1-2E)(1-E)}.$$

Grim. The Grim Strategy has two states: (C, P) (Cooperate, Punishment). The expected payoffs in the two states are, respectively:

$$\begin{aligned} V_C &= 2(G-1)(1-E) + \delta(V_C(1-E)^2 + (1-(1-E)^2)V_P), \\ V_P &= \frac{2E(G-1)}{1-\delta}. \end{aligned}$$

Solving for V_C we get:

$$V_C = 2(G-1) \frac{(1-\delta)(1-E) + E^2\delta(2-E)}{(1-\delta)(1-\delta + E\delta(2-E))}.$$

The IC constraint in the Punishment state is always satisfied since the players play an infinite repetition of the static Nash equilibrium. So to check under what conditions Grim is a PPE, we need to only check the IC constraint in the Cooperate state, which is

$$2G - 2E(1+G) + \delta((1-E)EV_C + (1-(1-E)E)V_P) \leq V_C.$$

It holds when:

$$\delta \geq \delta_{PPE, Grim} \equiv \frac{1}{(1-E)(G(1-2E) + E)}.$$

Threshold $\delta_{PPE, Grim}$ is less than 1 if and only if $G = \frac{1+g}{g} > \frac{1-E+E^2}{(1-2E)(1-E)}$, as claimed in part (i) of Proposition 2 (we show next that the threshold for Payback is lower, hence if δ is high enough for Grim to be a PPE, it is also high enough for Payback to be PPE).

Comparing the threshold factors for Grim and Payback Define $X \equiv (G(1-2E)(1-E) + E)$ and $A \equiv E(1-2E)(1-E)$. In this notation, we can write:

$$\begin{aligned}\delta_{PPE, Payback} &\equiv \frac{-X + \sqrt{X^2 + 4A}}{2A}, \\ \delta_{PPE, Grim} &= \frac{1}{X - E^2}.\end{aligned}$$

Let Z be the ratio of these thresholds:

$$Z \equiv \frac{\delta_{PPE, Payback}}{\delta_{PPE, Grim}} = \frac{(X - E^2)(-X + \sqrt{X^2 + 4A})}{2A}.$$

First, note that the limit of Z as $G \rightarrow \infty$ is 1:

$$\lim_{X \rightarrow +\infty} \frac{(X - E^2)(-X + \sqrt{X^2 + 4A})}{2A} = 1.$$

Using this limit, we show that $Z < 1$ for all parameters by showing that Z is increasing in G .

The derivative of the numerator of Z with respect to X is:

$$\begin{aligned}&\frac{\partial}{\partial X}((X - E^2)(-X + \sqrt{X^2 + 4A})) \\ &= -\frac{(2X - E^2)\sqrt{X^2 + 4A} - 2X^2 - 4A + E^2X}{\sqrt{X^2 + 4A}}.\end{aligned}$$

Note that this expression is zero at $A = 0$, and positive for all $A > 0$.

Therefore, for all parameters $\frac{\partial Z}{\partial X} > 0$, which implies that Z is strictly increasing in G . Hence, for all parameters, $Z < 1$. This establishes the rest of the claims in Proposition 2. $\square$

B.2.2 Proof of Proposition 3

Payback vs. Grim The expected PPE payoff for Payback is:

$$V_{Payback} = 2 \frac{(G-1)(1-E)}{(1-\delta)(1+E\delta(1-2E))}.$$

while for Grim:

$$V_{Grim} = 2(G-1) \frac{(1-\delta)(1-E) + E^2\delta(2-E)}{(1-\delta)(1-\delta + E\delta(2-E))}.$$

Therefore, to show $V_{Payback} > V_{Grim}$ we need to show that:

$$\frac{1 - E}{1 + E\delta(1 - 2E)} > \frac{(1 - \delta)(1 - E) + E^2\delta(2 - E)}{(1 - \delta + E\delta(2 - E))}.$$

An algebraic manipulation shows that this condition holds if and only if:

$$\delta(1 - E - 2E^2 + E^3) + 1 > 0.$$

For all $E < \frac{1}{2}$, this expression is positive, and that establishes the comparison between the equilibrium payoffs for Payback and Grim.

Payback vs. TFT If players follow TFT (starting in the Cooperative state), the expected payoff is:

$$V_{TFT} = 2(G - 1) \frac{1 - E + \delta(2E - 1)}{(1 - \delta)(1 + \delta(2E - 1))}.$$

Using our computation for $V_{Payback}$, we need to show that:

$$\frac{1 - E}{(1 + E\delta(1 - 2E))} > \frac{1 - E + \delta(2E - 1)}{1 + \delta(2E - 1)}.$$

After some algebra, this is equivalent to

$$\delta > -\frac{E^2}{E(1 - 2E)},$$

which is always satisfied since $\delta > 0$ and the r.h.s. is negative. $\square$

B.2.3 Proof of Proposition 4

We start with calculations for the size of the basin of attraction (BAD) for Payback, Grim and TFT when compared to AllD.

BAD for Payback. When I play Payback and meet another player that plays Payback, my expected payoff is

$$V_{Payback, Payback} = 2 \frac{(G - 1)(1 - E)}{(1 - \delta)(1 + E\delta(1 - 2E))}.$$

When I meet a player that plays AllD, my expected payoff is a solution to the system that describes expected payoffs in the three states of Payback:

$$\begin{aligned} U_C &= 2(E(1+G) - 1) + \delta((1-E)^2 U_P + 2E(1-E)U_C + E^2 U_R), \\ U_P &= 2E(G-1) + \delta((1-E)U_P + EU_C), \\ U_R &= 2(E(1+G) - 1) + \delta((1-E)U_C + EU_R). \end{aligned}$$

The solution for the expected payoff at the beginning of the game is:

$$V_{Payback, ALLD} = 2 \frac{\delta(1-E)^2(2\delta E^2 - \delta E + 1) - 2\delta E^2 G(1-E) + E(1+G) - 1}{(1-\delta)(1-2\delta E(1-E))}.$$

If instead I play AllD and I meet a player that plays AllD, I get a payoff:

$$V_{ALLD, ALLD} = \frac{2E(G-1)}{1-\delta}.$$

Finally, if I play AllD and I meet a player that plays Payback, I get a payoff that is a solution to the following system of equations (where subscripts now represent the state of Payback of the other player).

$$\begin{aligned} U_C &= 2G - 2E(1+G) + \delta((1-E)^2 U_P + 2E(1-E)U_C + E^2 U_R), \\ U_P &= 2E(G-1) + \delta((1-E)U_P + EU_C), \\ U_R &= 2G - 2E(1+G) + \delta((1-E)U_C + EU_R). \end{aligned}$$

Solving this system yields expected payoff at the beginning of the game:

$$V_{ALLD, Payback} = 2 \frac{G(1-\delta) - (1+G)E + EG(1-2E)(1-E)^2\delta^2 - E\delta(-2G + EG - 2E + 2E^2)}{(1-\delta)(1-2\delta E(1-E))}.$$

Let α be the fraction of other players using the Payback strategy. Player 1 is indifferent between AllD and Payback if

$$\alpha V_{Payback, Payback} + (1-\alpha)V_{Payback, ALLD} = \alpha V_{ALLD, Payback} + (1-\alpha)V_{ALLD, ALLD}.$$

Solving for α we get:

$$\alpha_{Payback}^* = \frac{(1-\delta E(1-E))(1+\delta E(1-2E))(1-\delta(1-E))}{(1-2E)(1-E)(1-\delta^2 E^2(1-E))(G-1)\delta}.$$

BAD for Grim. When I play Grim and meet another player that plays Grim, my expected payoff is

$$V_{Grim,Grim} = 2(G-1) \frac{(1-\delta)(1-E) + E^2\delta(2-E)}{(1-\delta)(1-\delta + E\delta(2-E))}.$$

When I play Grim and meet a player that plays AllD, my expected payoff is a solution to the system (where subscripts represent the two states of my Grim strategy):

$$\begin{aligned} U_C &= 2(E(1+G) - 1) + \delta((1-E)EU_C + (1 - (1-E)E)U_P) \\ U_P &= \frac{2E(G-1)}{1-\delta}, \end{aligned}$$

The resulting expected payoff is:

$$V_{Grim,ALLD} = 2 \frac{\delta E^2(G-1)(-1+E) + (G+1-2\delta)E - 1 + \delta}{(1-\delta)(1-\delta E + \delta E^2)}.$$

Finally, if I play AllD and I meet a player that plays Grim, I get a payoff that is a solution to the following system of equations (where subscripts represent now the states of the other player):

$$\begin{aligned} U_C &= 2G - 2E(1+G) + \delta((1-E)EU_C + (1 - (1-E)E)U_P), \\ U_P &= \frac{2E(G-1)}{1-\delta}. \end{aligned}$$

The resulting expected payoff is:

$$V_{ALLD,Grim} = 2 \frac{G(1-\delta) - \delta E^2(G-1)(1-E) - (1+G(1-2\delta))E}{(1-\delta)(-\delta E + \delta E^2 + 1)}.$$

Let α be the fraction of other players using the Grim strategy. Player 1 is indifferent between AllD and Grim if

$$\alpha V_{Grim,Grim} + (1-\alpha)V_{Grim,ALLD} = \alpha V_{ALLD,Grim} + (1-\alpha)V_{ALLD,ALLD}.$$

Solving for α we get:

$$\alpha_{Grim}^* = \frac{1-\delta + E\delta(2-E)}{\delta(1-2E)(1-E)(G-1)}.$$

BAD for TFT. When two players that play TFT meet, each has expected payoff of:

$$V_{TFT,TFT} = 2(G-1) \frac{1-E + \delta(2E-1)}{(1-\delta)(1+\delta(2E-1))}$$

When I play TFT and meet a player that plays AllD, my expected payoff is:

$$V_{TFT,ALLD} = \frac{2(E(G+1) - 1) + \delta(1 - E)(1 - 2E)}{1 - \delta}$$

If I play AllD and meet a player that plays TFT, I get expected payoff :

$$V_{ALLD,TFT} = \frac{2(G(1 - E)(1 + \delta(-1 + 2E)) - E)}{1 - \delta}$$

Let α be the fraction of other players using the TFT strategy. Player 1 is indifferent between AllD and TFT if

$$\alpha V_{TFT,TFT} + (1 - \alpha)V_{TFT,ALLD} = \alpha V_{ALLD,TFT} + (1 - \alpha)V_{ALLD,ALLD}.$$

Solving for α we get:

$$\alpha_{TFT}^* = \frac{1 - \delta + 2\delta E}{(G - 1)\delta(1 - 2E)}$$

Comparison: Grim vs. Payback Using our formulas for α_{Grim}^* and $\alpha_{Payback}^*$ we get that the difference is:

$$\begin{aligned} & \alpha_{Grim}^* - \alpha_{Payback}^* \\ = & E \frac{E^4\delta^2 + 2\delta(1 - \delta)E^3 + 1 - \delta E(1 + E(1 - \delta))}{(1 - 2E)(1 - E)(1 - \delta^2 E^2 + \delta^2 E^3)(G - 1)} \end{aligned}$$

The denominator is positive, and the numerator is positive because

$$1 - \delta E(1 + E(1 - \delta)) \geq 1 - E > 0.$$

Comparison: TFT vs. Payback We want to show that $\alpha_{Payback}^* < \alpha_{TFT}^*$, which is:

$$\frac{(1 - \delta E(1 - E))(1 + \delta E(1 - 2E))(1 - \delta(1 - E))}{(1 - E)(1 - \delta^2 E^2(1 - E))} < 1 - \delta(1 - 2E)$$

which can be rewritten as:

$$\frac{(1 - \delta E(1 - E))}{(1 - \delta^2 E^2(1 - E))} < \frac{(1 - \delta(1 - 2E))(1 - E)}{(1 + \delta E(1 - 2E))(1 - \delta(1 - E))}.$$

Since the l.h.s. is lower than 1, it is enough to show that the r.h.s. is greater than 1, which is equivalent to:

$$1 - \delta(1 - 2E) - E + E\delta(1 - 2E) > 1 + \delta E(1 - 2E) - \delta(1 - E)(1 + \delta E(1 - 2E)).$$

Collecting terms, this is equivalent to:

$$1 < 1 + (1 - E)E\delta,$$

which is true for any $(E, \delta) \in (0, \frac{1}{2}) \times (0, 1)$. $\square$

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Online appendix

C Formal justification for S-Payback

This appendix provides the formal analysis underlying the conditions under which Suspicious Payback (S-Payback) and Payback can jointly emerge in equilibrium. We consider a population composed of two types of players: conditional cooperators, who adopt either Payback or Suspicious Payback, and always-defectors (AD). Let θ denote the fraction of conditional cooperators in the population, and $1 - \theta$ the fraction of AD players.

Conditional cooperators choose whether to start a supergame by cooperating (playing Payback) or defecting (playing S-Payback). Let p represent the probability that a conditional cooperator plays Payback; thus, $1 - p$ is the probability of starting with S-Payback. The decision between these two strategies depends on the comparison of the expected continuation payoffs from cooperation (V_c) and defection (V_d) in the first round.

Using the normalized payoff matrix in Table B1, the expected payoff from cooperating in the first round is:

$$V_{Payback} = \frac{\theta p}{1 - \delta} - (1 - \theta p)l + \delta \theta (1 - p) \left((1 + g) + \frac{\delta}{1 - \delta} \right), \quad (2)$$

while the expected payoff from defecting in round 1 – as prescribed by S-Payback – is:

$$V_{S-Payback} = \theta p (1 + g - \delta l + \frac{\delta^2}{1 - \delta}). \quad (3)$$

In equilibrium, conditional cooperators must be indifferent between starting with cooperation or defection. Setting $V_{Payback} = V_{S-Payback}$ and solving for p yields:

$$p = \frac{l - l\delta + (-1 + g(-1 + \delta))\delta\theta}{-g + l + (-1 + g - l)\delta^2\theta}. \quad (4)$$

Since p must lie between 0 and 1, the fraction θ of conditional cooperators must fall within an admissible range, $[\underline{\theta}, \bar{\theta}]$, where:

$$\underline{\theta} = \min \left(\frac{l}{\delta - g + \delta l + l}, \frac{\delta l - l}{-\delta + \delta^2 g - \delta g} \right), \quad \bar{\theta} = \max \left(\frac{l}{\delta - g + \delta l + l}, \frac{\delta l - l}{-\delta + \delta^2 g - \delta g} \right)$$

When θ is below $\underline{\theta}$, cooperation in the first period is never optimal; when θ is above $\bar{\theta}$, cooperation is always optimal and Suspicious Payback does not arise.

In addition to characterizing the behavior of conditional cooperators, it is important to explain why some players might persist with Always Defect. We assume that players who adopt AD hold incorrect beliefs about the behavior of conditional cooperators: they mistakenly believe that cooperators play Grim trigger rather than Payback, and that no Suspicious Payback is present. Under such beliefs, they perceive the effective fraction of always-defectors as $(1 - \theta) + \theta(1 - p)$. Because such beliefs give no incentive to switch to cooperation if the opponent cooperates in round 1, these players can never learn about the existence of Payback and S-Payback players. Therefore, they do not update their beliefs and, consistent with the idea of a self-confirming equilibrium (Fudenberg and Levine, 1993), they continue to prefer defection.

An AD-player evaluating a deviation to Grim trigger, but holding wrong beliefs about the strategy adopted by the opponent, would expect a payoff of:

$$V'_{GT} = \frac{\theta p}{1 - \delta} - (1 - \theta p)l < V_{AD} = \theta p(1 + g) \quad (5)$$

This condition requires that

$$\theta p < \frac{l(1 - \delta)}{(l - g)(1 - \delta) + \delta}$$

Hence, Payback, S-Payback and Always Defect can coexist in equilibrium, provided that the AD-players hold incorrect – and sufficiently pessimistic – beliefs about the strategy adopted by conditional cooperators.

D Complementarities between strategies

To quantify the complementarity between Payback, S-Payback, and other strategies, we perform a computational exercise. First, we retrieve the MLE of strategies on the last 4 supergames of each session of the STANDARD dataset, excluding Payback and S-Payback, yielding 37% TFT, 20% S-TFT, 15% Grim, and 28% ALLD. Second, we compute the average coordination rate on mutual cooperation for each possible strategy pair and for different expected game durations (e.g., $\delta = 0.875$ implies an expected duration of 8 periods). For instance, when TFT and S-TFT are paired under perfect monitoring, their interaction alternates between dc and cd , resulting in a coordination rate of 0%. These rates are then weighted by the probability of each match and aggregated, yielding the overall coordination rate on mutual cooperation at the population level.

Figure D1 presents the analysis under perfect monitoring. The simulation results (left panel) reinforce the theoretical insights from Section 3. Under perfect monitoring, replacing Tit-for-Tat (TFT) with Payback (red line) leaves the coordination rate unchanged, as Payback does not generate complementarities with other strategies in the absence of noise.

In contrast, S-Payback strategy enhances the coordination rate when it replaces the S-TFT (blue line), enabling re-coordination on mutual cooperation from period 2 onward when matched with TFT (or Payback). As a result, reallocating 20% of the population share from S-TFT to S-Payback leads to (i) a higher average coordination rate that (ii) increases with the expected duration of the supergame. To empirically corroborate these findings, the right panel of Figure D1 displays observed coordination rates by expected game duration in the STANDARD dataset. The empirical estimates align closely with the simulated outcomes with S-Payback replacing S-TFT (blue line, left panel), both qualitatively and quantitatively, providing further support for the behavioral relevance of repayment-based strategies. Comparable patterns are also evident in analyses of average payoffs and average cooperation rates.

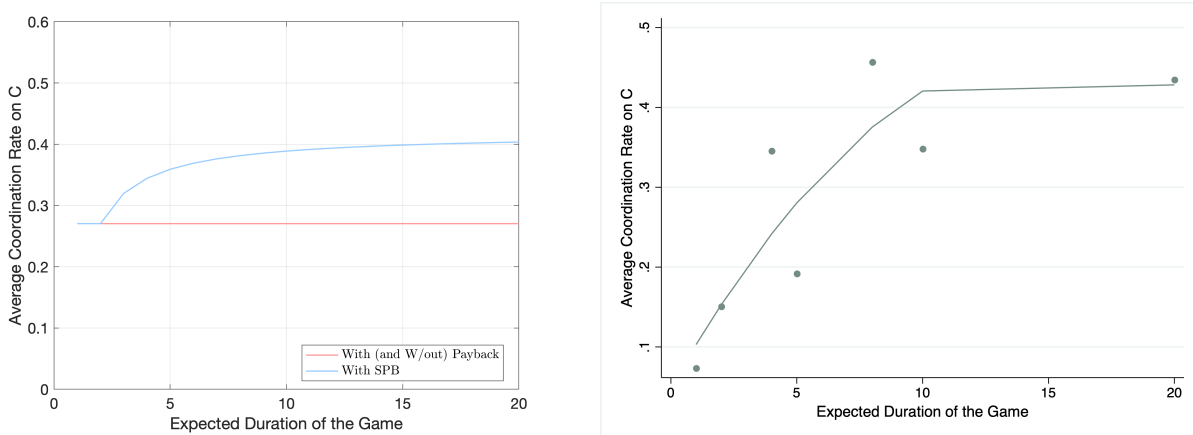


Figure D1: Average coordination rate on mutual cooperation as a function of the expected duration of the supergame under perfect monitoring. *Left panel:* Simulations. The red line shows both the baseline case — 37% TFT, 20% S-TFT, 15% Grim, and 28% ALLD — and the variant where TFT is replaced by Payback. The blue line only replaces S-TFT with S-Payback. *Right panel:* Empirical coordination rate by discount factor in the STANDARD dataset.

Figure D2 reports simulations for the imperfect monitoring case, with coordination rates averaged on 1,000 simulated supergames to account for stochastic realizations of the noise. To isolate the intensive margin of cooperation, and to facilitate a direct comparison with the left panel of figure D1, we retain the same strategy shares estimated under perfect monitoring. However, it is important to note that the introduction of noise typically changes the strategies used, favoring more lenient and forgiving strategies (see Fudenberg et al., 2012). Due to this shift—and the limited variability in the empirical data regarding the expected duration of supergames—we do not find meaningful to compare the simulations directly with observed cooperation rates. Instead, we focus on evaluating the impact of substituting TFT and S-TFT with Payback and S-Payback in simulations.

The results confirm again the points made in Section 3. Under imperfect monitoring, both Payback (red solid line) and S-Payback (blue solid line) have an independent positive impact on the average coordination rate compared to the baseline case (black solid line). When considered jointly (blue dashed line), the average coordination rate doubles. While the impact of Payback and S-Payback under imperfect monitoring could be reduced by the extensive presence of lenient strategies, that allow to re-coordinate starting from the second period after deviation, the theoretical properties and the computational analyses shed light on the crucial complementary role played solely by Payback and S-Payback. This is in line with findings A and B in Section 3, which show that repayment strategies emerge also in the NOISE dataset.

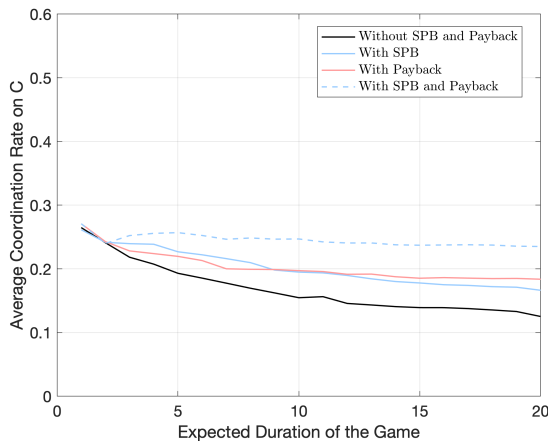


Figure D2: Average coordination rate on mutual cooperation as a function of the expected duration of the supergame under imperfect monitoring, with $E = \frac{1}{8}$. *Black solid line*: simulations using the baseline strategy frequencies (37% TFT, 20% S-TFT, 15% Grim, and 28% ALLD). *Red solid line*: replaces TFT with Payback, keeping S-TFT. *Blue solid line*: replaces S-TFT with S-Payback, keeping TFT. *Blue dashed line*: replaces both TFT and S-TFT with Payback and S-Payback, respectively.

E Strategic Risk under perfect monitoring

In this subsection, we estimate strategy frequencies in the STANDARD dataset, separately for treatments in which cooperation is a subgame-perfect Nash equilibrium but not risk-dominant, and for treatments in which it is. Following Fudenberg and Karreskog (2024), we further distinguish between cases where the discount factor δ lies just above the threshold for risk dominance ($\delta < \delta_{RD} + 0.3$), and those in which the continuation probability is well above the threshold.³⁰ As in Table 8, we analyze separately the supergames in which the

³⁰ Fudenberg and Karreskog (2024) notice that “there is more cross-session variation in average cooperation rates for intermediate values of δ_{RD} ” and we need “caution in interpreting experimental studies of

first action is cooperation (C) or defection (D). Results are reported in Table E1.

The first empirical pattern that emerges from Table E1 is a marked increase in aggregate cooperation in supergames that admit a risk-dominant equilibrium: average cooperation rates rise from 20% to 46% ($N_1 = 19$, $N_2 = 56$, p-value < 0.01) and from 46% to 75% ($N_1 = 56$, $N_2 = 21$, p-value < 0.01).³¹ A similar pattern holds for first-round actions, where the share of cooperative moves increases from 23% to 53% ($N_1 = 19$, $N_2 = 56$, p-value < 0.01) and from 53% to 83% ($N_1 = 56$, $N_2 = 21$, p-value < 0.01). This finding is consistent with previous results reported by Blonski et al. (2011) and Dal Bó and Fréchette (2018).

A second notable observation is that Payback does not appear to outperform other cooperative strategies in this setting: its estimated frequency is comparable to that of alternative approaches that also begin with cooperation. This result is consistent with Proposition 1, which shows that under perfect monitoring, the threshold discount factor for risk dominance is at least as high for Payback as it is for Grim trigger. Moreover, as previously noted, it is generally difficult to distinguish between Payback and TFT in these data. For many participants, the observed histories from the last four supergames do not allow us to tell these two strategies apart with confidence.

³¹ *prisoner's dilemmas with parameters in this range that are based on only a small number of sessions*". The statistical tests reported here and in Section 4.2.2 are performed using a Wilcoxon-Mann-Whitney test with session averages as the unit of observation.

Level of Strategic Risk	$\delta < \delta_{RD}$	$\delta_{RD} \leq \delta < \delta_{RD} + 0.3$	$\delta \geq \delta_{RD} + 0.3$
Cooperation Rate	20%	46%	75%
Cooperation Rate period 1	23%	53%	83%
Full			
Payback	4%	9%	25%
TFT	6%	15%	17%
S-Payback	7%	7%	2%
S-TFT	21%	10%	2%
AllD	51%	28%	9%
Grim	3%	16%	22%
Others	7%	14%	24%
<i>Observations</i>	5.640	25.454	7.726
<i>Subjects</i>	316	960	344
Begin by C			
Payback	15%	12%	22%
TFT	32%	35%	27%
Grim	24%	29%	25%
Others	28%	23%	26%
<i>Observations</i>	1.401	13.694	6.341
<i>Subjects</i>	117	636	305
Begin by D			
S-Payback	16%	20%	28%
S-TFT	21%	20%	13%
AllD	62%	59%	57%
<i>Observations</i>	4.239	11.760	1.385
<i>Subjects</i>	281	572	89

Notes: The analyses reported in this Table are based on the STANDARD dataset, restricted to treatments in which a positive level of cooperation can be sustained as a subgame-perfect Nash equilibrium. Statistics are based on the last four supergames of each session. All strategies grouped under the row “Others” begin with cooperation.

Table E1. Cooperation Rate and Strategy Adoption in STANDARD by Strategic Risk

A third key insight from Table E1 concerns the strategies adopted by participants who defect in the first round of a supergame. In this subset of the data, the estimated frequency of S-Payback rises from 16% when $\delta < \delta_{RD}$, to 20% when $\delta_{RD} \leq \delta < \delta_{RD} + 0.3$, and to 28% when $\delta \geq \delta_{RD} + 0.3$. This pattern should not be surprising. Consider a participant who begins a supergame with defection – whether due to pessimistic expectations, caution, or lack of trust – but then observes that the opponent cooperated in the first round. When cooperation is risk-dominant, the incentives to re-coordinate become particularly strong. Among the strategies available, S-Payback provides an attractive path to restoring mutual cooperation after such an initial misstep.³² To our knowledge, this is the first study to show

³² As noted in subsection 4.2, the pair (S-Payback, Payback) constitutes an SPE – or a PPE under

that risk dominance not only correlates with the prevalence of cooperation, but also enhances the appeal of specific strategies.

F Additional Tables

	Observations	Subjects	Supergames
STANDARD dataset			
All	39.784	1.734	6.922
Start D	18.264	1.052	3.406
Start C	21.520	1.077	3.516
Overlapping Subjects		395	
NOISE dataset			
All	20.322	598	2.206
Start D	7.211	286	795
Start C	13.111	430	1.411
Overlapping Subjects		118	
FINITE dataset			
All	13.144	552	2.148
Start D	4.994	327	1.011
Start C	8.150	363	1.137
Overlapping Subjects		138	

Notes: Last 4 supergames of each session, keeping all periods.

Table F1: Summary information on the meta-dataset of repeated PD experiments for MLE estimation.

imperfect monitoring – across all subgames that do not begin in the initial exploration state. By contrast, no such equilibrium can be sustained with the pair (S-TFT, TFT). This implies that as $\delta - \delta_{RD}$ increases also $\delta - \delta_{SPE, Payback}$ increases and, therefore, that S-Payback should become increasingly attractive as compared to S-TFT. Table F2 in Online Appendix F reports the results using $\delta - \delta_{SPE, Payback}$ and confirms this intuition.

	$\delta < \delta_{SPE,Grim}$	$\delta_{SPE,Grim} \leq \delta < \delta_{SPE,Payback}$	$\delta \geq \delta_{SPE,Payback}$
Cooperation Rate	7%	29%	51%
Cooperation Rate period 1	8%	32%	58%
Full			
Share Payback	0%	3%	13%
Share TFT	3%	11%	14%
Share S-Payback	0%	6%	6%
Share S-TFT	25%	22%	8%
Share AllD	69%	41%	26%
Share Grim	2%	4%	17%
Share Others	0%	12%	16%
Observations	964	5.046	33.774
Subjects	114	254	1.366
Start C			
Share Payback	32%	31%	13%
Share TFT	48%	20%	34%
Share Grim	15%	15%	29%
Share Others	5%	34%	23%
Observations	84	1.655	19.781
Subjects	19	113	945
Start D			
Share S-Payback	0%	16%	21%
Share S-TFT	22%	23%	19%
Share AllD	78%	61%	60%
Observations	880	3.391	13.993
Subjects	110	199	743

Notes: Estimates on the last 4 Supergames of each sessions, data from the STANDARD dataset

Table F2. Cooperation Rate and Strategy adoption under perfect monitoring with different levels $\delta_{SPE,Grim}$ and $\delta_{SPE,Payback}$

Remark on Table F2. The estimations is the first column of the table ($\delta < \delta_{SPE,Grim}$) for Start C are highly unstable due to the small number of subjects and the fact that no history can differentiate between TFT and Payback. Therefore the estimates vary widely and are not informative, since in this case Payback cannot be differentiated from TFT. The only information that is stable is that 68% use either Payback or TFT.

Treatment:	$g=l=2$ $E=1/8$	$g=l=1$ $E=1/8$	$g=l=2/3$ $E=1/8$	$g=l=1/3$ $E=1/8$	$g=l=1/3$ $E=1/16$	$g=l=1/3$ $E=0$
Strategies						
<i>V_{coop}</i>						
Payback	6.5	12.9	19.4	38.8	42.9	48
Grim	3.3	6.5	9.8	19.6	25.7	48
TFT	5.1	10.2	15.3	30.6	35.2	48
<i>δ_{PPE}</i>						
Payback	0.85	0.67	0.55	0.36	0.30	0.25
Grim	0.91	0.70	0.57	0.37	0.30	0.25
TFT	/	/	/	/	/	/
<i>δ_{RD}</i>						
Payback	0.96	0.83	0.73	0.54	0.46	0.4
Grim	>1	0.91	0.80	0.57	0.47	0.4
TFT	>1	0.89	0.76	0.53	0.46	0.4
<i>Size-BAD</i>						
Payback	0.81	0.40	0.27	0.13	0.08	0.05
Grim	>1	0.57	0.38	0.19	0.11	0.05
TFT	>1	0.52	0.35	0.17	0.10	0.05
<i>Cooperation rates</i>						
First Period (FRD)	54%	75%	79%	76%	87%	83%
First Period (Arechar et al.)	33%	50%	/	/	/	/
Overall (FRD)	32%	49%	61%	59%	82%	78%
Overall (Arechar et al.)	21%	37%	/	/	/	/

Notes: The table includes data from Fudenberg et al. (2012) and Arechar et al. (2017). The value of cooperation is obtained from Tables 4 and 6 in FRD, while the critical thresholds derive from our calculations. The cooperation rates are for the last 4 supergames.

Table F3. Theoretical benchmarks and observed cooperation under imperfect monitoring.